

Bevacizumab in Breast Cancer – Deep Research Report

Drug Summary

Bevacizumab is a humanized monoclonal antibody targeting vascular endothelial growth factor A (VEGF-A), thereby inhibiting tumor angiogenesis ¹. In breast cancer, bevacizumab has been evaluated primarily in HER2-negative metastatic disease, combined with chemotherapy. These trials showed significant improvement in progression-free survival (PFS) and response rates (doubling median PFS from ~5–6 months to ~9–11 months) ², particularly in aggressive subtypes like triple-negative breast cancer (TNBC) which are often highly angiogenic. However, no significant overall survival (OS) benefit was demonstrated (OS hazard ratio ~1.0) ³. This modest benefit, coupled with increased serious adverse events, led the US FDA to withdraw approval of bevacizumab for metastatic breast cancer in 2011 ⁴ ⁵. In contrast, the European Medicines Agency (EMA) maintained its authorization in combination with paclitaxel (and later capecitabine) for selected patients, acknowledging a PFS benefit while emphasizing the need to identify responders ⁶.

Clinical use of bevacizumab in breast cancer is now limited to specific scenarios. It may be considered for **TNBC or other HER2-negative patients with high tumor burden or visceral crisis**, where a rapid tumor shrinkage could palliate symptoms ⁷. Notably, retrospective analyses suggest TNBC patients derive a greater PFS benefit (HR ~0.63 for PFS) than hormone receptor-positive patients ³. No definitive predictive biomarker is established, but baseline VEGF-A levels have been explored: high circulating VEGF-A was associated with better PFS outcomes in trials ⁸ ⁹. Bevacizumab is not incorporated into standard U.S. guidelines for breast cancer due to the lack of OS improvement, but European and some international guidelines allow its use in **selected cases** with careful risk-benefit consideration ⁷. Emerging research also highlights bevacizumab's impact on the tumor microenvironment – for example, **short-term VEGF blockade can normalize tumor vasculature and increase immune cell infiltration**, providing a rationale for combining bevacizumab with immunotherapies in TNBC ¹⁰. Overall, bevacizumab's role in breast cancer remains as an adjunct for aggressive disease subtypes, with its efficacy largely contingent on tumor angiogenic dependency and balanced against significant safety considerations.

Identifiers & Synonyms

- **ChEMBL ID:** ChEMBL1201583 ¹¹
- **DrugBank ID:** DB00112 ¹²
- **Synonyms / Brand Names / Codes:** Bevacizumab is marketed under the brand name **Avastin** (originator). Numerous biosimilars and international trade names include *Abevmy*, *Alymsys*, *Avzivi*, *Aybintio*, *Mvasi* (bevacizumab-awwb), *Zirabev* (bevacizumab-bvzr), *Vegzelma*, *Oyavas*, and others ¹³. Development and biosimilar codes for bevacizumab have included **rhuMAB-VEGF** (original recombinant humanized anti-VEGF antibody) ¹⁴ as well as identifiers like **ABP-215**, **BAT1706**, **CT-P16**, **BP102**, **IBI305**, **MIL60**, **PF-06439535**, etc., corresponding to various investigational or biosimilar products ¹⁵. These all refer to the same antibody molecule or its close copies.

Mechanism of Action (Breast Cancer Context)

Bevacizumab exerts its anti-cancer effect by binding to **VEGF-A** and preventing this key pro-angiogenic ligand from engaging its receptors (VEGFR-1 and VEGFR-2) on endothelial cells. In the context of breast cancer, many tumors – especially more aggressive subtypes – overexpress VEGF in response to hypoxic stress via HIF-1 α activation. **By neutralizing circulating VEGF-A, bevacizumab blunts downstream VEGF signaling**, leading to pruning of immature tumor blood vessels and normalization of the remaining vasculature. This transient “vascular normalization” improves oxygenation and delivery of concurrently administered chemotherapy to tumor tissue, partially explaining the improved response rates and PFS seen when bevacizumab is added to chemo in metastatic breast cancer ².

Within the breast tumor microenvironment, VEGF-driven angiogenesis creates aberrant, leaky vessels that foster hypoxia and immune evasion. Bevacizumab counteracts this by temporarily restoring more orderly blood flow. Notably, VEGF also has immunomodulatory effects: it can inhibit dendritic cell maturation and promote an immunosuppressive milieu. VEGF blockade has been shown to **enhance immune cell infiltration** in breast tumors – for example, a single dose of bevacizumab in primary TNBC patients led to a reduction in Ang2⁺ dysfunctional vessels and a marked increase in intratumoral CD4⁺ and CD8⁺ T cells and mature dendritic cells ¹⁰. This suggests that in VEGF-rich, “immune-cold” tumors (often TNBC), bevacizumab may help jump-start an anti-tumor immune response by normalizing vasculature and relieving VEGF-mediated immunosuppression.

However, the **benefit of VEGF inhibition tends to be cytostatic and context-dependent**. Tumors that are highly **angiogenesis-dependent** (e.g. some basal-like TNBCs) initially respond with slowed growth or regression due to starvation of blood supply. In contrast, more indolent luminal tumors or those that have alternative blood vessel supply mechanisms show less responsiveness ¹⁶. Furthermore, breast cancers can adapt: chronic VEGF blockade leads to increased intratumoral hypoxia, which **triggers escape pathways** and selection for more invasive, VEGF-independent cancer cell clones ¹⁷ ¹⁸. This manifests as upregulation of compensatory pro-angiogenic factors (like FGF2, ANG2) and activation of epithelial-mesenchymal transition (EMT) and metastasis programs, ultimately limiting bevacizumab’s effectiveness (see **Resistance** sections). Thus, while bevacizumab’s mechanism—angiogenesis inhibition—remains consistent across breast cancer subtypes, its **impact is greatest in tumors that are VEGF-driven (e.g. many TNBCs)** and can be transient as tumors invoke resistance mechanisms. Combination strategies (with chemotherapy, immune checkpoint inhibitors, or other pathway blockers) are being explored to prolong and deepen responses by offsetting these adaptive resistance pathways ¹⁰ ¹⁹.

Primary Targets (Human)

- **VEGFA (vascular endothelial growth factor A):** This is the principal and direct molecular target of bevacizumab. Bevacizumab binds all major isoforms of human VEGF-A with high affinity, preventing VEGF-A from activating VEGF receptors ¹. By sequestering VEGF-A, bevacizumab effectively inhibits the VEGF/VEGFR signaling axis responsible for new blood vessel formation. VEGFA is a secreted growth factor; in breast cancer it is often produced by tumor cells and tumor-associated stromal cells under hypoxic conditions, making it a critical driver of angiogenesis. Bevacizumab’s clinical activity is derived from this **specific neutralization of VEGFA**, which leads to downstream effects on tumor vasculature and the microenvironment as described above. (No other direct human protein targets are known for bevacizumab, as it does not bind VEGF-B, VEGF-C, or the VEGF receptors themselves – its action is specifically through VEGF-A ligand blockade.)

Pathways (Overview)

Bevacizumab's influence in breast cancer spans **multiple biological pathways and processes**, primarily centered on angiogenesis and the tumor microenvironment. Below is an overview of key pathways affected, with an indication of whether these pathways are upregulated or downregulated in the context of bevacizumab treatment and whether they contribute to drug sensitivity or resistance:

- **Angiogenesis / VEGF Signaling (MSigDB HALLMARK_ANGIOGENESIS, KEGG hsa04370):** *Downregulated by bevacizumab; associated with drug sensitivity.* This is the core pathway targeted – VEGF-driven blood vessel formation. Bevacizumab sharply reduces pro-angiogenic signaling by VEGF-A neutralization. Tumors highly reliant on angiogenesis (high baseline VEGF pathway activity) are more sensitive to bevacizumab ⁹, as the drug deprives the tumor of its neovasculature. Inhibition of angiogenesis leads to transient vascular normalization and tumor starvation, a desired effect that underlies initial responses.
- **Hypoxia / HIF-1α Pathway (GO:0001666 response to hypoxia, HALLMARK_HYPOXIA):** *Upregulated (as an adaptive response); associated with resistance.* By pruning the tumor's blood vessels, bevacizumab often **exacerbates intratumoral hypoxia**, stabilizing HIF-1α and activating hypoxia-responsive genes ¹⁷. This hypoxic shift is a double-edged sword: it can slow tumor growth initially, but it also induces a cascade of survival and escape mechanisms. Hypoxia-driven pathways promote angiogenic rebound (e.g. increased VEGF production, induction of alternative angiogenic factors) and more aggressive behavior, thereby undermining long-term efficacy of bevacizumab ¹⁸.
- **Epithelial–Mesenchymal Transition (EMT) & Invasion (MSigDB HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION):** *Upregulated in response to VEGF blockade; contributes to resistance.* Anti-angiogenic therapy's hypoxia stress can push cancer cells to adopt a mesenchymal, invasive phenotype. Studies have observed **upregulation of EMT transcription factors** (Twist1, Snail) after anti-VEGF treatment, loss of E-cadherin, and gain of vimentin, corresponding to increased motility and invasiveness ²⁰. This EMT activation enables tumor cells to disseminate and find new blood supplies, a mechanism of **adaptive resistance** and possibly increased metastasis after prolonged bevacizumab exposure ²¹. (Notably, preclinical models showed enhanced invasiveness and metastasis in some settings of VEGF or VEGFR inhibition ²², although this has not been clearly proven in clinical breast cancer.)
- **Cancer Stem Cell and Basal-Like Phenotype (e.g. ALDH+ stemness, basal gene expression):** *Upregulated with resistance.* Bevacizumab pressure can enrich for a subpopulation of cancer cells with stem-like properties. In an ER-positive xenograft model, VEGF blockade caused a shift from a luminal phenotype to a more **basal-like, stem-cell-associated phenotype** as resistance developed ²³. Genes associated with cancer stem cells were elevated in bevacizumab-resistant tumors ²⁴. This suggests that the therapy may selectively favor a quiescent, stem-like cell population that can survive low-angiogenesis conditions and drive rebound growth once therapy-induced vascular suppression wanes.
- **Notch Signaling Pathway (Reactome R-HSA-157118, “Signaling by NOTCH”):** *Upregulated in resistant tumors.* Notch is a key pathway in cell fate and stem cell maintenance. Gene expression analyses have identified **Notch pathway activation as a top correlate of bevacizumab resistance** in breast cancer models ²⁵. Notch signaling in tumor cells and the vasculature (via endothelial cell–

tumor cell crosstalk) can promote angiogenic independence and support the cancer stem cell compartment. Thus, heightened Notch activity has been linked to reduced bevacizumab efficacy, and there is interest in Notch inhibitors to potentially overcome this mechanism of resistance.

- **Fibroblast Growth Factor (FGF) / Fibroblast Growth Factor Receptor (FGFR) Pathway (MSigDB hallmark or KEGG hsa04090):** *Upregulated as a compensatory angiogenic route; associated with resistance.* The FGF-FGFR axis is an alternate pro-angiogenic signaling pathway. In the setting of VEGF blockade, tumors can increase production of FGF2 (basic FGF) and related factors. **Upregulation of FGF2 has been correlated with resistance to anti-VEGF therapy** in several tumors ²⁶. FGF2 can stimulate angiogenesis by recruiting pericytes and activating FGFR on endothelial and stromal cells, effectively bypassing the VEGF pathway. In breast cancer, preclinical evidence shows that high FGF2 (for example, driven by tumor or obesity-related factors) can blunt bevacizumab's effect, whereas dual inhibition of VEGF and FGF signaling yields more potent anti-tumor effects ²⁷ ¹⁹.
- **Angiopoietin/Tie2 Pathway (e.g. ANGPT1/2-Tie2 signaling, Reactome R-HSA-210993):** *Upregulated in escape; resistance-associated (when active).* Angiopoietin-2 (Ang2) is a pro-angiogenic factor that destabilizes blood vessels and works in concert with VEGF. Bevacizumab does not target Ang1/Ang2, and evidence suggests that **tumors may upregulate Ang2 during anti-VEGF treatment** as a compensatory mechanism. High serum Ang2 has been linked to poor bevacizumab responses in other cancers ²⁸. In breast cancer, a beneficial effect of bevacizumab was to **reduce Ang2-positive vessel density** (signifying pruning of abnormal vessels) ¹⁰. If Ang2 remains high, it can recruit pro-angiogenic myeloid cells and facilitate vessel regrowth, as well as contribute to immunosuppression. Thus, insufficient suppression or rebound of the Ang/Tie2 pathway (Ang2 up) is considered a resistance pathway. Combination approaches blocking both VEGF and Ang2 (e.g. with bispecific antibodies) have shown promise in preclinical models ²⁹.
- **HGF/c-MET Signaling Pathway (KEGG hsa04010 or HALLMARK_HGF_SIGNALING):** *Upregulated under hypoxia; contributes to resistance.* The hepatocyte growth factor (HGF) and its receptor c-MET drive cell migration, growth, and angiogenesis. Anti-VEGF-induced hypoxia can directly increase c-MET expression and activation via HIF-1 α ³⁰ ³¹. The HGF/MET pathway has been strongly implicated in tumors that escape VEGF inhibition: for instance, bevacizumab-treated tumors (notably in glioblastoma, and potentially in basal breast cancers) often show **upregulation of c-MET**, which promotes invasive growth and new vessel formation through VEGF-independent means ³¹ ³². This has led to trials combining bevacizumab with MET inhibitors to counteract this resistance mechanism. While specific data in breast cancer are limited, MET overactivation is plausibly relevant in aggressive, hypoxic breast tumors under VEGF blockade.
- **Immune Response and Tumor Immune Microenvironment (Immune gene signatures, e.g. MSigDB C7 IMMUNE_RESPONSE):** *Can be upregulated (improved) by bevacizumab, which is a sensitivity-enhancing effect.* Bevacizumab's vascular effects modulate immune cell trafficking into tumors. By normalizing vessels, bevacizumab can **increase infiltration of effector T-cells and dendritic cells**, as demonstrated in TNBC patients receiving neoadjuvant bevacizumab ¹⁰. An **"immune-hot" microenvironment** (T-cell inflamed) is generally associated with better therapeutic response; thus, bevacizumab's ability to convert an immune-cold tumor to a more immune-permissive state may enhance sensitivity, especially in combination with immunotherapies. On the other hand, certain inflammatory pathways, if chronically upregulated, can be detrimental (see IL6

below). Overall, a shift toward an immune-active tumor microenvironment is a positive pathway change linked to bevacizumab sensitivity.

- **IL-6 / Inflammatory Signaling Pathway (GO:0038156 interleukin-6 signaling):** *Upregulated in special contexts (e.g. obesity); associated with resistance.* Chronic inflammation and high IL-6 levels can undermine bevacizumab's efficacy. Obesity is one scenario: obese breast cancer patients have elevated systemic IL-6 and FGF-2, larger tumor burden, and showed poorer responses to anti-VEGF therapy ³³ ¹⁹. IL-6 promotes angiogenesis and immunosuppressive inflammation. Preclinical studies found that blocking IL-6 in obese mouse models restored sensitivity to bevacizumab, indicating that an **IL-6-driven inflammatory pathway can mediate resistance** ³⁴ ¹⁹. Thus, while not a universal mechanism, an upregulated IL-6 pathway represents a subtype-specific resistance factor (e.g. in patients with metabolic inflammation). Targeting IL-6 or related cytokines may enhance bevacizumab effects in such settings.

(The above are the major pathways identified. Other pathways like PDGF signaling, Integrin signaling, and VEGF-C/D-mediated lymphangiogenesis also play supporting roles in the angiogenic network and resistance but are omitted here for brevity. The subsequent sections provide more detailed, context-specific analysis of these pathway alterations.)

Upregulated Pathways (Post-Bevacizumab Exposure)

Certain pathways have been observed to become **upregulated (heightened in activity or expression)** in the context of bevacizumab treatment or resistance in breast cancer. These upregulated pathways can represent the tumor's adaptive responses and may contribute to either sensitivity or resistance depending on their nature:

- **Hypoxia/HIF-1 Pathway:** As noted, VEGF blockade often leads to increased tumor hypoxia. Markers of hypoxia (e.g. HIF1A, CA9) become upregulated after bevacizumab exposure ¹⁷. In breast tumor models, pimonidazole staining (hypoxia indicator) rose following anti-VEGF therapy, confirming that **bevacizumab-upregulated hypoxia** is a real outcome ³⁵. This upregulation is generally detrimental – HIF-1α activation underlies many downstream resistance mechanisms (e.g. inducing VEGF again, EMT, MET, etc.). Thus, while hypoxia is a direct result of successful angiogenesis inhibition (initially a sign of drug activity), its **sustained upregulation is a hallmark of emerging resistance**.
- **EMT and Invasion Pathways:** Following from hypoxia, pathways associated with EMT, cell migration, and invasion become upregulated. Gene expression changes in bevacizumab-treated tumors include **increases in transcription factors like TWIST1, SNAI1/SNAI2**, and mesenchymal markers (N-cadherin, vimentin) ¹⁸. In patient-derived xenografts and cell line models, long-term bevacizumab led to more infiltrative growth patterns. Clinically, this upregulation may manifest as more locally invasive disease or diffuse recurrence patterns after bevacizumab. It suggests that **pathways enabling cells to move and survive in a nutrient-poor environment (EMT, matrix remodeling, integrin signaling)** are activated as a compensatory response.
- **Notch Signaling:** Upregulation of Notch pathway components (e.g. NOTCH1, JAG1, HES1) has been linked to bevacizumab resistance. In experimental ER+ breast tumors engineered to overexpress VEGF, **Notch pathway genes were elevated in tumors that acquired resistance** to bevacizumab

²⁵ . Notch upregulation can occur in both tumor cells and tumor endothelial cells under VEGF blockade, promoting cell survival and new vessel sprouting via non-VEGF angiogenesis. This is an **adaptive upregulation** that likely supports the maintenance of a cancer stem cell population and vessel co-option, enabling tumor regrowth despite VEGF suppression.

- **Follistatin/Activin Pathway:** Interestingly, the same gene expression study identified **FST (follistatin)** as one of the top upregulated genes in bevacizumab-resistant tumors ²⁵ . Follistatin is an antagonist of Activin (a TGF- β family member). Upregulation of FST could reflect a complex feedback: possibly the tumor is altering TGF- β /Activin signaling in response to VEGF inhibition. The net effect might be promotion of a pro-proliferative environment (since Activin can be growth-inhibitory, its binding by FST might relieve a brake on tumor cells). While the exact role is unclear, this highlights that **TGF- β superfamily signaling changes** (e.g. Activin, BMP pathways) are part of the adaptive landscape – some of which might encourage angiogenic escape or EMT.
- **Alternate Angiogenic Factors:** Bevacizumab selectively targets VEGF-A, so tumors may upregulate alternate pro-angiogenic ligands. Key among these are:
 - **FGF2 (basic fibroblast growth factor):** Often found to increase in bevacizumab-treated tumors ²⁶ . FGF2 can drive angiogenesis independent of VEGF and recruit supporting stromal cells (pericytes). In breast cancer models, elevated FGF2 has been directly linked to resistance; conversely, combining FGFR inhibitors or FGF traps with anti-VEGF restored tumor control ²⁷ ¹⁹ .
 - **Angiopoietin-2:** As discussed, Ang2 is frequently upregulated during anti-angiogenic therapy. Clinically, **high Ang2** is associated with poor outcome on bevacizumab ²⁸ . Ang2 upregulation destabilizes vessels but in the absence of VEGF can paradoxically facilitate **vessel co-option and protection** of the vasculature by recruiting peri-endothelial cells. Some breast tumors under bevacizumab show rising Ang2 expression as they become more invasive.
 - **Placental Growth Factor (PlGF):** PlGF (VEGF family member) can bind VEGFR-1. It is often low in breast cancer, but studies note **PlGF upregulation in resistant tumors** ³⁶ ³⁷ . PlGF can stimulate angiogenesis and inflammation without using VEGFR-2, thus bypassing bevacizumab's effect. This has led to interest in agents like aflibercept (which also neutralizes PlGF).
 - **VEGF-C / VEGF-D:** These drive lymphangiogenesis via VEGFR-3, and to some extent angiogenesis via VEGFR-2. Under VEGF-A blockade, some tumors increase VEGF-C/D, potentially promoting lymphatic vessel growth and alternate vascularization ³⁸ ³⁶ .
- **Inflammatory Cytokines (IL-6, IL-8):** Inflammatory pathways can be paradoxically upregulated in response to VEGF blockade. For instance, hypoxia in tumors stimulates **IL-8 (CXCL8)**, a pro-angiogenic chemokine that can attract neutrophils and endothelial precursor cells. Elevated IL-8 has been noted in some models of anti-angiogenic resistance ³⁹ . Similarly, as mentioned, **IL-6 is elevated in certain patient subsets (obesity)** and contributes to an upregulated inflammatory state that counteracts vessel normalization ³³ ¹⁹ . These cytokine pathways create a microenvironment that supports angiogenesis via alternate routes and tumor cell survival, reducing bevacizumab efficacy.

In summary, **upregulated pathways during or after bevacizumab treatment** include those related to hypoxic stress (HIF1 α), cellular plasticity and invasion (EMT, Notch, MET), and compensatory angiogenic signals (FGF2, Ang2, cytokines). These changes are largely implicated in **resistance**, as the tumor attempts

to circumvent VEGF-A blockade. Recognizing these upregulated signals provides opportunities for combination therapies (for example, adding MET inhibitors, FGFR inhibitors, or anti-IL-6 antibodies alongside bevacizumab to forestall resistance).

Downregulated Pathways

Bevacizumab's action also leads to **downregulation or suppression of certain pathways**, either directly through its mechanism or indirectly as a consequence of reduced tumor perfusion. Understanding which pathways are downregulated can reveal how bevacizumab creates vulnerabilities in the tumor. Key downregulated pathways and processes in breast cancer include:

- **Angiogenesis/VEGF Pathway:** This is the intended downregulation – **VEGF signaling is markedly suppressed** by bevacizumab. The transcription of VEGF-regulated endothelial genes (like those involved in new vessel sprouting) is reduced. Microarray data from treated tumors show decreased expression of many pro-angiogenic factors and their receptors compared to untreated tumors ⁴⁰. In clinical breast tumors, bevacizumab can lead to a **drop in tumor microvessel density** and blood flow. For example, dynamic imaging studies have shown reduced perfusion after bevacizumab, reflecting the downregulation/collapse of the tumor's neovasculature. In primary tumor analyses, VEGF downstream genes (like ANGPT2, KDR/VEGFR2) were found downregulated in many breast cancers relative to normal tissue – a finding which suggests some tumors naturally suppress angiogenesis as they grow ⁴¹. Bevacizumab further enforces this anti-angiogenic state, essentially **shutting down the VEGF-VEGFR2 axis**. This downregulation is beneficial (required for antitumor effect), but as noted, tumors may adapt around it.
- **Tumor Perfusion and Metabolic Activity:** With angiogenesis inhibited, pathways related to **aerobic metabolism and cell proliferation** tend to be downregulated. Well-oxygenated tumors typically have active oxidative phosphorylation and cell-cycle progression. After VEGF blockade, breast cancer models often show **downregulation of genes involved in cell cycle (cyclins) and DNA replication**, consistent with growth arrest. Clinical studies using PET imaging demonstrated decreased glucose uptake in tumors after bevacizumab, indicating a downshift in metabolic activity. These effects suggest that bevacizumab can **downregulate proliferation pathways** indirectly by inducing nutrient deprivation. For instance, mTOR signaling (nutrient-sensing pathway) might be attenuated due to lower oxygen and growth factor availability, though specific gene evidence in breast tumors is limited.
- **Estrogen/Hormone Signaling (in certain contexts):** An interesting observation from resistant models: tumors that shifted to a basal phenotype under bevacizumab had **downregulation of luminal estrogen receptor (ER) signaling pathways** ⁴². Essentially, the ER-positive, luminal gene expression program was lost (ER-dependent genes downregulated) as the tumor became more basal/stem-like. While this was a consequence of resistance, it highlights that **bevacizumab pressure can downregulate hormone-driven pathways** in some ER+ cancers, possibly by selecting ER-negative subclones or causing cells to abandon energy-intensive differentiation programs. Clinically, this could mean that an ER+ tumor responding to bevacizumab might behave more like an ER-negative one over time. However, in general, **hormonal pathways are not a primary direct target** of bevacizumab; any downregulation is indirect and context-specific.

- **Lymphangiogenesis Pathways:** VEGF-A also has some overlap with lymphatic vessel growth (and VEGF-C/-D share some receptors). There is evidence that **bevacizumab can downregulate lymphangiogenesis** in tumors to a degree. Markers like VEGF-C, VEGF-D, and **LYVE-1 (lymphatic endothelial marker)** were found decreased in some breast tumors vs. normal tissue and are presumably suppressed further when VEGF-A is blocked ⁴⁰. This is logical because reducing blood vessel growth often also impacts lymphatic expansion (tumors often co-opt both). Reduction in lymphangiogenic activity might transiently slow nodal metastasis, although this has not been clearly proven in trials.
- **Anti-Angiogenic Feedback Molecules:** Paradoxically, some natural angiogenesis inhibitors produced by the body can be **upregulated** as a feedback to tumor angiogenesis (e.g., thrombospondin-1, endostatin). In highly angiogenic tumors, these endogenous inhibitors are often suppressed. Bevacizumab treatment, by reducing VEGF, may relieve negative feedback and allow anti-angiogenic factors to rise. In one study, **thrombospondin-1 (TSP1)** was found elevated (less suppressed) in bevacizumab-responsive tumors, effectively a **downregulation of pro-angiogenic bias** and tilt towards angiogenesis suppression. This could be viewed as “downregulation” of pathways that degrade the extracellular matrix for angiogenesis or recruit endothelial cells. While not frequently quantified, it’s part of the picture that bevacizumab can restore some of the body’s natural angiogenesis breaks.
- **Immune Checkpoint Signals (potentially):** There is some evidence (mostly preclinical) that by normalizing vessels and reducing certain cytokines, bevacizumab might **downregulate immunosuppressive pathways**. For example, **VEGF can induce PD-L1 expression** on macrophages and dendritic cells; blocking VEGF might reduce this immunosuppressive signal, effectively downregulating a checkpoint pathway. Additionally, bevacizumab has been associated with a drop in regulatory T-cell infiltration in some studies (though results are mixed). If present, these would represent downregulation of **immunosuppressive pathways** (e.g. Treg recruitment, certain IL-10 or VEGF-mediated PD1/PD-L1 interactions), thereby sensitizing the immune response against the tumor.

In summary, **downregulated pathways due to bevacizumab** primarily revolve around **angiogenesis and proliferation** – the drug successfully knocks down the tumor’s ability to form new vessels and can induce a broad slowing of growth signals. This is the desired effect and correlates with **tumor sensitivity** to the drug (tumors where these pathways are effectively downregulated tend to respond). However, these very changes set the stage for the tumor to mount countermeasures (which often show up as the upregulated pathways discussed earlier). The balance between sustained pathway downregulation (good for tumor control) and emergent pathway upregulation (driving resistance) determines the long-term efficacy of bevacizumab in breast cancer.

Sensitivity Pathways (Factors Increasing Sensitivity)

This section highlights biological pathways or biomarkers whose presence or upregulation **increases a breast cancer’s sensitivity to bevacizumab**. In essence, these are contexts where the tumor is more vulnerable to VEGF inhibition. Evidence for sensitivity mechanisms in breast cancer includes:

- **High VEGF/Angiogenesis Signature (VEGF-High Tumors):** Tumors that are inherently **highly angiogenesis-driven** tend to be more sensitive to bevacizumab. Clinical analyses across trials have

suggested that patients with elevated baseline **plasma VEGF-A** derive greater PFS benefit from bevacizumab ⁹. Likewise, in retrospective tissue studies, tumors with high microvessel density and high VEGF expression responded better (greater tumor shrinkage) to anti-VEGF therapy. This makes intuitive sense: a **“VEGF-addicted” tumor** will be crippled when VEGF is taken away. In practical terms, this sensitivity pathway is seen in some **basal-like and TNBC tumors**, which often overexpress VEGF and have hypoxia-inducible angiogenic drive. These tumors experience significant growth inhibition from bevacizumab due to pruning of their extensive neovasculature.

- **Angiogenic Profile (VEGF:PIGF, VEGFR2 levels):** Beyond VEGF-A itself, certain angiogenic biomarkers correlate with sensitivity. For example, the AVADO trial's biomarker study found that patients with higher baseline **VEGFR-2** levels in plasma also had more benefit ⁹. One hypothesis is that tumors causing an upregulation of soluble VEGFR-2 reflect a highly angiogenic milieu that is especially susceptible to VEGF blockade. A low level of decoy receptor sVEGFR-1 (which soaks up VEGF) was also associated with better bevacizumab effect ⁴³ – presumably because more free VEGF was available and then neutralized by the drug. In summary, a **pro-angiogenic profile** (high pro-angiogenic factors, low natural anti-angiogenic factors) tilts a tumor toward being **bevacizumab-sensitive**.
- **Immune-Inflamed Microenvironment:** Emerging evidence suggests that tumors with a pre-existing immune response might respond better to VEGF inhibition. VEGF can impair dendritic cell function and T-cell infiltration, so tumors that are on the cusp of immune activation could be “unlocked” by bevacizumab. In a subset analysis of a neoadjuvant trial, **patients whose tumors had higher baseline T-cell infiltration (TILs)** experienced higher pathologic complete response rates when bevacizumab was added to chemotherapy, particularly in TNBC (though results vary in different studies). Moreover, as noted, bevacizumab can *create* a more immune-friendly tumor bed ¹⁰. Thus, an **immune-hot tumor (e.g. TNBC with lymphocytic infiltration)** represents a context of sensitivity – VEGF blockade will aid immune cells to penetrate and attack, complementing chemotherapy. This is one rationale behind combining bevacizumab with immunotherapies: tumors sensitive to bevacizumab often show immune activation that can be further exploited.
- **BRCA1/2 Mutation or HRD (hypothesis):** While not conclusively proven, there has been speculation that **BRCA-mutated or homologous recombination-deficient (HRD)** breast cancers (many of which are TNBC) might be more sensitive to anti-angiogenic therapy. The logic is that BRCA/HRD tumors tend to be more genomically unstable, possibly more hypoxic (due to rapid growth), and they often have higher neoangiogenesis and immune cell infiltration. Some retrospective analyses of trials like LEA or RIBBON-2 hinted that patients with triple-negative, BRCA-mutated tumors had improved outcomes with bevacizumab, though patient numbers were small. This remains an area of investigation. If true, the pathway reasoning is that **defective DNA repair and resultant genomic instability** lead to an **“angry” tumor microenvironment (immune activation and VEGF upregulation)**, which in turn makes the tumor highly susceptible to VEGF blockade. However, currently this is not an established clinical biomarker for bevacizumab; it's a theoretical sensitivity factor.
- **Low Alternative Angiogenic Signals:** Tumors that **lack strong alternative angiogenic pathways** are more likely to be sensitive to bevacizumab because they cannot easily compensate for VEGF inhibition. For instance, a cancer that does *not* have upregulated FGF2, Ang-2, or PDGF pathways at baseline will be essentially “trapped” when VEGF is blocked, as it has no backup way to grow blood

vessels. In contrast, a tumor that already has multiple angiogenic drivers can switch gears. As a concrete example: in one study, responders to bevacizumab had **lower baseline FGF2 levels** than non-responders ³³ ¹⁹. Similarly, patients with obesity (who often have elevated FGF2 and IL-6) responded less, implying that *not* having those factors (i.e. lean patients with lower inflammatory angiogenic factors) conferred sensitivity. So, a **clean VEGF-dominated angiogenesis program** (without parallel pathways active) is a sensitizing context.

- **Normalization of Vasculature (Transient):** Once bevacizumab is given, tumors that undergo **effective vascular normalization** appear to respond better. This is less a baseline pathway and more a dynamic response: in imaging studies, patients whose tumor blood flow initially decreased (indicating vessel pruning) but then stabilized at a lower, more efficient level were the ones with longer PFS on bevacizumab. This suggests that if the tumor's angiogenic pathway can be tamped down to a manageable level (rather than completely obliterating vessels or no effect at all), chemotherapy delivery and oxygenation improve – a phenomenon known as the “normalization window.” Pathways related to **pericyte coverage and vessel maturity** are involved here. Tumors that readily **downregulate abnormal vessel growth and upregulate vessel maturation** in response to VEGF blockade (for example, via increased Ang-1/Tie2 signaling relative to Ang-2) likely have better outcomes. In essence, a tumor environment conducive to normalization (as opposed to one that immediately collapses into extreme hypoxia) is more sensitive.
- **Subtype Context – Triple-Negative vs Luminal:** Clinically, **TNBC is the subtype most often associated with bevacizumab sensitivity**. TNBCs frequently have high proliferation, high VEGF, and immune cell infiltration – all factors mentioned above. Pooled analyses confirm that TNBC patients see a larger PFS improvement than hormone receptor-positive patients with bevacizumab ³, even though OS is unchanged. On the other hand, **luminal A (ER+/HER2- low-grade) tumors**, which rely more on hormonal pathways and have relatively low microvessel density ¹⁶, often do not benefit much – they are essentially angiogenesis-“cold” and thus bevacizumab-insensitive. This dichotomy underscores that the **pathways driving TNBC (angiogenesis, immune activation)** align with bevacizumab's mechanism, whereas the pathways in luminal cancers (ER signaling, endocrine drivers) do not. Therefore, being a **basal-like or TNBC tumor** can be considered a proxy for harboring the sensitivity pathways (like high angiogenic drive), whereas **being a luminal tumor indicates absence of those pathways** and relative insensitivity.
- **Specific Biomarkers in Trials:** Researchers have explored specific genes or signatures. For example, one study identified an “angiogenesis signature” of 26 genes that predicted bevacizumab benefit in ER+ breast cancer patients ⁴⁴. If a tumor had that signature upregulated, it was sensitive. Another analysis found that **low levels of vascular normalization inhibitors** (like SDF1 or VEGF-C) in the tumor correlated with better responses, as the remaining vessels could normalize more easily. While these are not used clinically, they highlight that **fine-tuned pathway balance** – high pro-angiogenic vs low anti-angiogenic factor ratio – yields sensitivity.

In summary, breast cancers are most sensitive to bevacizumab when **angiogenesis is a dominant survival pathway** for the tumor. The pathways reflecting this state (VEGF high, alternative angiogenesis low, active immune infiltration, high microvascular density) serve as hallmarks of bevacizumab sensitivity. These conditions are frequently met in aggressive triple-negative tumors. Recognizing a “VEGF-addicted” phenotype can inform clinicians that bevacizumab is more likely to have meaningful efficacy (as was seen in

subsets of clinical trials), whereas its benefit will be modest to nil in tumors that do not heavily leverage VEGF for growth.

Sensitivity Pathways for Upregulation

(Pathways whose upregulation correlates with increased sensitivity to bevacizumab in breast cancer.)

- **VEGF/Angiogenesis Pathway Upregulation:** As discussed, tumors with **upregulated VEGF signaling** are more sensitive. An elevated **HALLMARK_ANGIOGENESIS** gene set or high VEGFA levels in the tumor microenvironment is a key indicator. Upregulation of this pathway means the tumor is highly dependent on VEGF-driven angiogenesis. Bevacizumab's efficacy is greatest in this context because blocking VEGF directly strikes at a critical vulnerability. For example, patients with **VEGF-A^{high}** status showed improved outcomes on bevacizumab ⁹. In these cases, the **biological rationale** is straightforward: the tumor has "all its eggs" in the VEGF basket for blood supply, so removing VEGF starves the tumor. This sensitivity has been observed most in TNBC, where VEGF pathway genes can be upregulated (sometimes via HIF1α in hypoxic, rapidly growing tumors) – leading to dramatic initial responses when VEGF is neutralized.
- **Immune Activation Pathways (Upregulated Immune Signatures):** Unexpectedly, the presence of an **upregulated immune response** within the tumor correlates with bevacizumab sensitivity. For instance, tumors with high expression of **CXCL9/CXCL10 (Th1-type chemokines), PD-1/PD-L1, or T-cell receptor activation genes** often indicate an immune-infiltrated tumor. These "hot" tumors respond better to bevacizumab because VEGF suppression further promotes immune cell activity (VEGF normally inhibits dendritic cells and T-cells). In one analysis, a high **Immune Metagene Signature** (reflecting upregulated immune pathways) was associated with improved PFS on bevacizumab in HER2-negative breast cancer ¹⁰. The *rationale* is that such tumors benefit doubly: bevacizumab prunes vessels to cut off tumor nutrition and also **enhances immune-mediated tumor killing** by relieving VEGF-induced immunosuppression. Thus, pathways like **Interferon-γ response, T-cell receptor signaling, and Mature dendritic cell signatures** – when upregulated – signify a milieu where bevacizumab can unleash an anti-tumor immune effect (especially when combined with chemotherapy or checkpoint blockade).
- **High Angiogenic Ligand/Receptor Pairs:** Upregulation of certain ligand–receptor pathways within angiogenesis indicates sensitivity. For example, **VEGFA-VEGFR2 axis** upregulation (high VEGFA and high KDR (VEGFR2) expression in tumor endothelial cells) means the tumor endothelium is actively driven by VEGF. Bevacizumab will cause a pronounced effect here – shutting down that communication. Similarly, upregulation of **VEGFR1 (FLT1)** on macrophages or stromal cells along with high VEGF could mark a VEGF-dependent stroma that will be affected by bevacizumab. In a sense, this overlaps with the general VEGF-high theme, but specifically, when both ligand and receptors are up, the whole pathway signal is strong and sensitive to interruption.
- **Hypoxia-Inducible Angiogenic Switch (short-term):** Counterintuitively, a tumor that has **upregulated hypoxia pathways** *before* treatment might be more sensitive initially. This is because it's already teetering on the edge of oxygen deprivation, heavily expressing VEGF in response – an "angiogenic switch" fully flipped on. Bevacizumab in such a scenario can cause an acute collapse of vasculature in the tumor leading to significant tumor cell death (an effect akin to what's seen in highly angiogenic tumors like renal cell carcinoma). For example, if a breast tumor has high HIF1α

and high VEGF (indicating upregulated hypoxia response driving angiogenesis), the initial VEGF blockade can induce a fatal blow to the tumor vessels. The caveat is that this can also shorten the normalization window and possibly speed resistance via even more hypoxia; however, in terms of immediate sensitivity, these tumors respond with quick shrinkage.

- **Lack of Redundant Pathways (implicitly, upregulation of one dominant pathway means others are not up):** This is a subtle point – if a tumor shows **strong upregulation of the VEGF pathway but not FGF, PDGF, etc.**, it implies VEGF is the dominant player. It's not exactly a pathway "upregulated" for sensitivity, but rather an isolated dependency. In practical terms, this might be identified by a molecular analysis showing VEGF and its targets high, while FGF2, ANGPT2, EGFR, etc., are not overexpressed. The tumor is a one-trick pony (VEGF-driven), and bevacizumab exploits that. This concept underlies efforts to define a "VEGF-dependent signature" for sensitivity.
- **Microenvironment features indicating angiogenic drive:** Upregulation of **endothelial cell proliferation genes, matrix metalloproteinases (MMPs) for vessel remodeling, and endothelial-specific markers (CD34, VWF)** can collectively signal an ongoing angiogenic process. When these are high, bevacizumab sensitivity is high. For instance, tumors with high **CD34 vessel density and elevated MMP9** (which is released by tip cells during angiogenesis) responded better in certain analyses. These are surrogates for an active angiogenic pathway upregulation that bevacizumab can shut down.

In essence, **Sensitivity Pathways for Upregulation** boil down to the tumor being heavily invested in making blood vessels and possibly engaging the immune system. Upregulation of angiogenesis and certain immune elements sets the tumor up for a **bigger fall when VEGF is blocked**. These contexts produce the most pronounced and durable responses to bevacizumab in breast cancer.

Sensitivity Pathways for Downregulation

(Pathways whose downregulation or low activity enhances sensitivity to bevacizumab.)

While most sensitivity factors involve something being high (e.g., high VEGF making the tumor vulnerable), there are cases where the **absence or suppression of a pathway** makes a tumor more dependent on VEGF, hence more sensitive to its inhibition. These are somewhat the flip side of resistance pathways:

- **Absence of Alternative Angiogenic Pathways:** Downregulation (or inherently low expression) of non-VEGF angiogenic factors like **FGF2, PDGF-B, Ang-1/2, EGFR** in a tumor means VEGF is the primary angiogenic driver. In effect, the tumor has put "all its eggs in one basket." For example, a breast tumor with **low FGF2 and low Ang2** cannot readily form vessels if VEGF is taken away. Clinical observations support this: patients whose tumors had **low baseline FGF2** levels were more likely to benefit from bevacizumab ¹⁹. Similarly, low **PIGF** (a VEGF family cytokine) might indicate lack of VEGFR-1 mediated angiogenesis, again leaving VEGF-A as the key player. Thus, **downregulation of compensatory angiogenic pathways** equates to heightened bevacizumab sensitivity. This concept suggests a biomarker strategy: measure alternate angiogenic factors – if they are low, the patient is a good candidate for VEGF inhibition.
- **Downregulation of VEGF Traps/Decoys:** The body has natural VEGF inhibitors like soluble VEGFR-1 (sFLT1) and thrombospondin. If a tumor environment has these highly expressed, some VEGF is

already neutralized endogenously, and adding bevacizumab yields diminishing returns. In contrast, if those pathways are downregulated (i.e., **low sVEGFR-1, low TSP1, low other anti-angiogenic molecules**), then the tumor's VEGF is fully active and a single agent like bevacizumab can make a big difference. Evidence: in an AVADO sub-analysis, low plasma sVEGFR1 was correlated with a better PFS benefit ⁴³. We can think of this as: **downregulation of the body's natural VEGF sinks** means the tumor is more "VEGF-replete," and bevacizumab's artificial VEGF sink will have a large impact.

- **Immunosuppressive Pathways Downregulated:** Tumors that naturally have **downregulation of immunosuppressive signals** may be more responsive to bevacizumab since there is less immune inhibition to overcome once vessels normalize. For instance, a tumor with **low TGF- β or low regulatory T-cell infiltration** (which are immunosuppressive) is more likely to mount an immune response when VEGF is blocked. In such a tumor, bevacizumab doesn't have to fight against a strong immunosuppressive environment, so the sensitivity is higher. One could say **downregulation of TGF- β , IL-10, or checkpoint molecules** (like if a tumor is PD-L1 negative but still has T-cells around) might predict a more robust effect from the immune stimulation aspect of bevacizumab. This is a softer criterion and hasn't been fully validated, but conceptually, **a tumor already inclined to immune attack (i.e., lacking strong immunosuppression)** will be a better target for any therapy that can boost immune infiltration.
- **ER Signaling Down (in ER+ tumors):** Among ER-positive breast cancers, those with lower ER pathway activity might derive slightly more benefit from bevacizumab. The rationale is that highly endocrine-driven tumors (ER pathway up) rely less on angiogenesis, whereas if an ER+ tumor has lost some reliance on ER (ER pathway downregulated), it might be shifting to more growth-factor-driven angiogenesis. In other words, **luminal B tumors** (more proliferative, often with higher VEGF) vs **luminal A tumors** (less proliferative, angiogenically quiescent). A luminal tumor with low ER-related gene expression (downregulation of the estrogen pathway) often correlates with higher grade and more angiogenesis – thus more bevacizumab sensitivity. This was indirectly observed: in the MERiDiAN trial population (HER2- metastatic), a trend suggested that patients with non-"luminal-A" profiles (higher proliferation, effectively less ER dominance) got more PFS benefit from bevacizumab (though this was not definitive). So, while not a primary factor, one could consider **downregulation of the estrogen signaling pathway** (or conversely, an **upregulation of proliferation gene sets like Ki-67, which usually means lower relative ER signaling**) as a marker of a tumor that might lean on angiogenesis and thus be sensitive to VEGF inhibition.
- **Downregulation of MET or IL-6 Pathways (when present):** If a tumor does not have certain escape pathways active at baseline, it is more vulnerable. For instance, if a tumor naturally has **low HGF/MET signaling** (perhaps low MET expression), it cannot easily compensate via MET upregulation when VEGF is blocked, making the initial impact of bevacizumab more lethal. Similarly, a tumor with **low IL-6 and low inflammatory cytokines** (e.g., in a young, lean patient with little systemic inflammation) will respond better since it won't immediately counteract bevacizumab with inflammation-induced angiogenesis. Essentially, **downregulated escape mechanisms** = no immediate Plan B for the tumor = higher sensitivity. Some of these factors (MET, IL-6, FGF) overlap with what we discussed as resistance when upregulated; here we are stating the corollary: if they start off down, that favors sensitivity.

It's important to note that identifying "sensitivity pathways for downregulation" is challenging in practice, because one often defines sensitivity by the absence of resistance factors. Clinical research tends to focus

on what presence correlates with benefit rather than what absence does. In the context of bevacizumab, the most actionable finding has been *low alternative proangiogenic factors* = good. For example, the **absence (downregulation) of FGF2 up front** is one such finding ¹⁹.

In summary, pathways that, when **quenched or inherently low**, make a tumor **extra sensitive** to bevacizumab include any alternative angiogenic or survival routes. The tumor is then highly **“VEGF-addicted” and unprotected**, so VEGF blockade hits it hard. A simplistic way to frame it: **the fewer workarounds a tumor has (all those workarounds being effectively off/downregulated), the more a single hit on VEGF will hurt it.**

(In the case of insufficient or conflicting evidence for specific “downregulation sensitivity” pathways in breast cancer, it is often safer to rely on the positive indicators of sensitivity described earlier. Indeed, most clinical biomarkers studied for bevacizumab have been additive factors (like VEGF-A, VEGFR-2 levels) rather than subtractive ones.)

Resistance Pathways (Mechanisms of Resistance)

Despite initial responses in some breast cancers, bevacizumab's benefit is often temporary. Tumors find ways to **resist VEGF-A inhibition** through diverse pathways. This section covers proven mechanisms of resistance, including genetic, signaling, and microenvironmental adaptations that enable tumor regrowth or treatment failure:

- **Upregulation of Alternative Pro-angiogenic Factors:** A primary resistance mechanism is that tumors simply switch on other angiogenesis pathways to compensate for VEGF-A blockade. As detailed earlier, common factors include:
 - **FGF2 (Fibroblast Growth Factor 2):** Many bevacizumab-resistant tumors show increased FGF2 expression ²⁶. FGF2 can stimulate angiogenesis via FGFR on endothelial cells and recruit pericytes via PDGF. In breast cancer models, resistance was associated with a several-fold rise in FGF2, and combining anti-VEGF with anti-FGF restored tumor control ²⁷.
 - **Angiopoietin-2:** Upregulation of Ang2 in the setting of VEGF inhibition leads to vessel remodeling that can favor either pruning or regrowth depending on context. In resistant disease, Ang2 often works with inflammatory cells to revascularize the tumor. High Ang2 was associated with progression on bevacizumab in some analyses ²⁹. This has prompted trials of Ang2 inhibitors plus bevacizumab.
 - **EG-VEGF (Endocrine gland-derived VEGF) and others:** Less studied in breast cancer, but tumors might invoke factors like **EGF, HGF, PDGF-BB, VEGF-C/D** to help either co-opt existing vasculature or form new vessels through different routes (e.g., lymphangiogenesis or vessel mimicry).
- **Redundancy in vessel supply:** Some tumors resist by increasing **vessel co-option** – instead of growing new vessels, they hijack existing normal blood vessels. This isn't a “molecular pathway” per se, but it's a behavior: seen especially in tissues like the lungs or brain metastases, where the tumor cells migrate along normal vessels. In breast cancer, increased invasiveness can lead to such co-option, effectively bypassing the need for VEGF angiogenesis and rendering bevacizumab moot.
- **Genomic Alterations and Bypass Signaling:** Tumor genetic changes can confer resistance:

- There isn't a classic single gene mutation that causes bevacizumab resistance akin to, say, KRAS mutation causing EGFR-inhibitor resistance (note: some exploratory analyses in colorectal cancer suggested high VEGF-C or VEGF-D levels, but in breast cancer no such singular genomic marker is validated). However, **amplification or overexpression of MET** has been noted as tumors become more aggressive under hypoxia ³¹ ³² . If a subclone with MET amplification exists, VEGF inhibition might favor its outgrowth, since MET can drive angiogenesis and cell survival independent of VEGF.
- **PTEN loss / PI3K activation:** Loss of the tumor suppressor PTEN leads to high HIF1 α and VEGF. Initially, that makes the tumor VEGF-driven (sensitive), but simultaneously PTEN loss activates the PI3K/AKT pathway strongly. There is some evidence that tumors with PTEN loss can be less sustainably responsive to bevacizumab – they might shrink, but the robust PI3K signaling allows survival in low-angiogenesis conditions and quick rebound via vascular mimicry. Along these lines, **PIK3CA mutations** (found in many HR+ breast cancers) didn't clearly predict resistance or sensitivity in trials, but high AKT activity could theoretically help a tumor survive anti-VEGF stress.
- **Endothelial cell mutations:** Though not reported in breast cancer specifically, resistance could in theory arise if the tumor-associated endothelial cells themselves acquired changes making them VEGF-independent. For example, an endothelial cell Ras mutation might drive angiogenesis from the inside. This has not been documented in patients, but it's a theoretical resistance at the vessel level.
- **Tumor Cell Invasive/Metastatic Switch:** As noted under EMT, a major resistance mode is that tumor cells become more invasive and migrate away to find or form new vasculature elsewhere. In models, **bevacizumab-resistant breast tumors were more infiltrative** and spread along tissue planes. Clinical correlation is seen in patterns like diffuse liver metastases or invasive local recurrences after bevacizumab. Mechanistically, this involves upregulating pathways for motility:
- **MET and AXL** (receptor tyrosine kinases) can be upregulated, aiding migration.
- **Integrin β 1** is induced by hypoxia and was found up in bevacizumab-resistant GBM; in breast cancer, increased integrin signaling can help cells latch onto blood vessels (co-option) and resist apoptosis ⁴⁵ .
- **EMT transcription factors** (Twist, Snail as mentioned) essentially reprogram cells to a motile, survival-prone state. These factors are a direct response to the hostile microenvironment created by VEGF blockade (high TGF- β , hypoxia).
- **Matrix remodeling enzymes** like MMPs are often elevated, helping carve paths through tissue. Overall, this “invasive switch” does not allow the tumor to grow in the original site under bevacizumab, but it lets it find new territory with blood supply – thus disease progresses via new metastases or diffuse infiltration.
- **Myeloid Cell Infiltration and Pro-angiogenic Inflammation:** Resistance is also **mediated by bone marrow-derived cells** that tumors recruit under VEGF blockade:
- **Tumor-Associated Macrophages (TAMs):** Paradoxically, VEGF blockade can increase certain TAM populations. Hypoxia triggers tumors to secrete **CXCL12 (SDF1)** and **CSF-1**, which recruit macrophages. These macrophages often express compensatory factors like Ang2, BV8, or MMP9 that aid in revascularization. In breast cancer models, higher infiltration of CD11b+ myeloid cells was noted at progression on bevacizumab. These cells physically assist in forming vessel-like structures or guiding endothelial cells.

- **Granulocyte-Colony Stimulating Factor (G-CSF):** Upregulation of G-CSF has been observed (e.g., in some pancreatic models, but relevant broadly). G-CSF brings neutrophils that can also spur angiogenesis via Bv8 (Prok2). In breast cancer, if a tumor starts secreting more G-CSF during therapy, it could be a sign of this inflammatory angiogenic rescue.
- **Regulatory T-cells and MDSCs:** While bevacizumab can improve immune infiltration, some tumors may counter by attracting more regulatory immune elements. An increase in **myeloid-derived suppressor cells (MDSCs)**, which often express VEGFR as well as alternative angiogenic factors, can support tumor vasculature and suppress cytotoxic immunity, helping the tumor survive.

Therapeutically, these resistance mechanisms point toward combination treatments: for instance, CSF1-R inhibitors to block macrophage recruitment or CXCR4 inhibitors to block SDF1 might extend bevacizumab effectiveness (strategies being tested in preclinical studies).

- **Metabolic Adaptations:** A subtle form of resistance is when tumor cells adapt their metabolism to the low blood supply. By switching to more efficient glycolysis and autophagy, they can survive in nutrient-poor, hypoxic conditions. Studies have shown anti-angiogenic treatment leads to **upregulation of autophagy genes** and sometimes a dependency on autophagy for survival under treatment. This isn't a "pathway" often measured clinically, but if a tumor robustly activates AMPK and autophagy (perhaps indicated by LC3B, BNIP3 upregulation under hypoxia), it can resist cell death and wait out the therapy. There is interest in whether autophagy inhibitors (like hydroxychloroquine) could be paired with bevacizumab to thwart this aspect of resistance.

- **Subtype-Specific Resistance Patterns:**

- In **HER2-positive** breast cancer, bevacizumab is not commonly used (due to HER2-targeted therapy being primary), but if used, one might find resistance through the dominant HER2/PI3K pathway continuing to drive tumor growth regardless of angiogenesis. So a HER2+ tumor might not "care" about VEGF as much – intrinsic resistance.
- In **ER-positive**/luminal tumors, as mentioned, these often have de-emphasized angiogenesis and rely on ER and PI3K. If bevacizumab is applied, such a tumor may not initially respond (de novo resistance) because it can sustain itself via its microvasculature which was already minimal or via co-opting vessels in hormone-rich microenvironments. Also, ER+ tumors might ramp up **VEGF-D** (promoting lymphatics) or simply progress slowly, outlasting bevacizumab's impact.
- In **TNBC**, which often responds initially, acquired resistance tends to follow the prototypical pathways we've listed: FGF2 up, MET up, EMT activation, etc. One interesting TNBC-specific finding: in some TNBC xenografts, long-term bevacizumab led to enrichment of a **CD44⁺/CD24⁻ stem-like population** (tying to the stemness pathway) and these cells expressed higher IL-8 and other factors, driving rebound angiogenesis.

Summarizing, resistance to bevacizumab in breast cancer is typically **multi-factorial** and involves a network of changes. The tumor essentially "learns" to grow despite angiogenesis inhibition by: 1. **Finding new ways to get blood (alternate angiogenic factors or vessel co-option).** 2. **Changing itself to need less blood (metabolic shift, invasion to oxygen-rich areas, entering dormancy/stem state).** 3. **Recruiting help from the host (macrophages, other cells that rebuild vasculature).**

Each of these corresponds to a pathway or set of pathways we can target. For instance, Notch and MET inhibitors might prevent the stem/invasion route, FGFR or Ang2 inhibitors tackle alternate vessels, and immune therapies or TKI combinations might handle the microenvironment aspects.

Resistance Pathways for Upregulation

(Pathways whose upregulation is associated with resistance to bevacizumab in breast cancer, by enabling escape from VEGF inhibition.)

- **FGF2/FGFR Pathway Upregulation:** One of the clearest resistance indicators is the **upregulation of the FGF2-FGFR signaling pathway**. When this pathway is ramped up, the tumor can promote angiogenesis without VEGF. Breast cancers that became resistant often showed rising FGF2 levels ²⁶. Upregulated FGFR signaling in endothelial and stromal cells leads to new vessel formation and pericyte recruitment even in the presence of bevacizumab. *How it causes resistance:* FGF2 directly stimulates endothelial cell proliferation (via MAPK and PI3K pathways) and also increases PDGF-B release to recruit pericytes, stabilizing vessels ⁴⁶ ⁴⁷. This essentially **bypasses the need for VEGF**, nullifying bevacizumab's effect. Clinically, tumors with elevated FGF2 (for instance, obese patients' tumors had high FGF2 and resisted therapy ³³ ¹⁹) do poorly on bevacizumab. Thus, **FGF2 pathway upregulation = resistance**, and FGFR inhibitors can restore sensitivity (as shown in xenografts where FGFR blockade plus bevacizumab was synergistic ⁴⁸).
- **Angiopoietin-2/Tie2 Upregulation:** When the **Ang2/Tie2 pathway is upregulated**, resistance is promoted. *Mechanism:* Angiopoietin-2, when high, antagonizes Ang1 and destabilizes vessels, but in a VEGF-inhibited environment, Ang2 then works with inflammatory cells to create abnormal vascular loops or help tumor cells co-opt vessels. High Ang2 is correlated with poor outcomes on bevacizumab ²⁸. In resistant tumors, Ang2 often goes up (sometimes secreted by hypoxic tumor cells or by infiltrating TAMs). This upregulation leads to the formation of **Tie2-expressing macrophages (TEMs)** that directly assist blood vessel sprouting in a VEGF-independent manner ²⁹. Essentially, upregulated Ang2 signals a pro-angiogenic switch that bevacizumab cannot inhibit. Targeting Ang2 (e.g., trebananib, a peptibody against Ang1/2) has shown an ability to delay resistance in other cancers, supporting that notion that **Ang2 up = bevacizumab resistance pathway**.
- **c-MET/HGF Pathway Upregulation:** Hypoxia-driven **upregulation of HGF and c-MET** is a potent resistance route ³⁰ ³¹. *Mechanism:* HGF (scatter factor) activation of c-MET on cancer cells triggers a program of invasion, EMT, and survival, as well as pro-angiogenic effects via upregulating VEGF (ironic in presence of bevacizumab) and other factors. Studies in breast and other cancers found that after anti-VEGF therapy, **c-MET was overexpressed** and tumor cells became highly invasive ³². In GBM patients, bevacizumab resistance was associated with MET upregulation at relapse ⁴⁹; similarly, aggressive basal breast cancers often have HGF in the stroma that can kick in when VEGF is blocked. Upregulated MET signaling allows tumor cells to migrate to find vessels (metastasis promotion) and also directly confers resistance to apoptosis. This pathway's role is so central that combining a MET inhibitor (like cabozantinib) with anti-VEGF showed improved outcomes in renal cancer ⁵⁰ and is being considered in breast cancer brain metastases. **Thus, c-MET upregulation is a clear resistance pathway:** it bypasses the need for local angiogenesis by enabling dispersal and it creates an alternate growth signal.
- **Notch Pathway Upregulation:** As described before, resistant tumors often have **Notch signaling up**. *Mechanism of resistance:* Upregulated Notch in tumor endothelium leads to a persistent quasi-stable vasculature that is VEGF-independent (since Delta-like ligand 4 (DLL4)/Notch signaling in endothelial cells actually prunes excessive sprouting and can induce a normalized but resistant

vessel phenotype). In tumor cells, Notch upregulation helps maintain stemness and EMT, allowing survival in low-angiogenesis conditions. The gene expression evidence showed **Notch pathway components elevated** in bevacizumab-resistant xenografts ²⁵. Therapeutically, there is proof of concept that inhibiting DLL4/Notch can overcome resistance to VEGF inhibitors, but toxicity is an issue. The bottom line: when Notch is turned **up**, the tumor is often shifting to a mode where it doesn't rely on angiogenic sprouting (thus VEGF inhibition is less relevant) and is focusing on self-renewal and co-option. Therefore, **Notch upregulation = bypass of bevacizumab effect**.

- **IL-6/STAT3 Inflammatory Pathway Upregulation:** If the tumor microenvironment responds to VEGF blockade with an **inflammatory reaction – especially IL-6/STAT3 upregulation – it can drive resistance**. *Mechanism:* IL-6 is a cytokine that triggers STAT3 signaling in both tumor cells and stromal cells, promoting survival, angiogenesis (via induction of VEGF and FGF, ironically), and immunosuppression. Upregulated IL-6 was identified as a key mediator of obesity-associated bevacizumab resistance ³³ ¹⁹. High IL-6 levels can mobilize MDSCs and also directly cause endothelial cells to resist apoptosis. STAT3 activation leads to expression of pro-angiogenic genes and anti-apoptotic factors. In essence, if a tumor cranks up IL-6 in response to bevacizumab, it ignites an alternative pro-growth pathway that undermines the therapy. This has been seen in ER+ obese mouse models: blocking IL-6 restored responsiveness to bevacizumab ³⁴. So, **IL-6/STAT3 upregulation is a resistance pathway**, especially in inflammation-rich contexts.
- **VEGF Rebound (re-upregulation of VEGF-A or family):** Surprisingly, some tumors may counter bevacizumab by simply **overexpressing VEGF-A even more** or other family members (VEGF-C, -D). For instance, post-bevacizumab biopsies sometimes show **increased VEGF mRNA** – the tumor senses the VEGF is being neutralized and pushes production (via HIF1, etc.). While the drug will bind a lot of it, if the levels get very high, some VEGF may escape binding or maintain some minimal signaling to keep vessels alive. Also, **VEGF-C** upregulation can activate VEGFR-2/3 pathways to compensate partially. This is a direct negative feedback loop: **upregulation of the very target (VEGF)**, attempting to outcompete the drug, is a resistance tactic. In practice, the drug dose may become insufficient relative to tumor VEGF output. This concept is akin to how tumor cells amplify a target oncogene to resist a targeted drug. Clinically, there isn't an easy way to overcome this except increasing drug (limited by toxicity) or better drug delivery, but it's notable.
- **Vascular Mimicry and Co-option Pathways:** When tumor cells start to form vessel-like channels themselves (vascular mimicry) or co-opt host vessels, certain pathways are upregulated:
- **Extracellular matrix remodeling (e.g., Laminin, Collagen IV, Integrin signaling)** are up: this helps tumor cells align into tube-like structures.
- **Stem cell and embryonic pathways** like VE-cadherin expression on tumor cells (observed in aggressive melanoma and possibly TNBC) allow tumor clusters to conduct blood.
- Essentially, the tumor circumvents angiogenesis by upregulating a program to create vessel mimicry. This is seen as a resistance mechanism in some histologic analyses: bevacizumab-treated tumors had regions of **PAS-positive loops** indicating vascular mimicry channels. Pathways like **Wnt signaling** might also be involved here (as Wnt can contribute to tumor plasticity).

Upregulated resistance pathways often overlap and reinforce each other – for example, hypoxia (up) triggers HIF1 (up) which triggers MET, Ang2, VEGF-C (all up), plus EMT (Twist up), plus IL-8 (up). So, a bevacizumab-resistant tumor can have a “*symphony*” of upregulated factors orchestrating escape.

Resistance Pathways for Downregulation

(Pathways whose downregulation leads to or signifies resistance to bevacizumab, often by removing constraints on tumor adaptation.)

While much of resistance comes from things turning on, there are also pathways that, when turned **off or lost**, contribute to bevacizumab resistance:

- **Downregulation of VEGF Dependency (Angiogenic Dormancy):** Some breast tumors inherently have low angiogenic drive (as noted in the 2009 BJC study: primary tumors often had **downregulation of angiogenesis factors compared to normal tissue** ⁴¹). Such tumors survive in a state of **angiogenic dormancy** – they don't need to actively sprout vessels. This can be considered a form of resistance *de novo*: if a tumor has effectively downregulated its need for VEGF (e.g., via selection of clones that thrive in low vessel density), bevacizumab has little impact. Therefore, one could say **"downregulation of angiogenesis itself" is a resistance state** – it sounds odd, but it means the tumor is no longer reliant on the targeted pathway (VEGF) and thus resistant. Clinically, this manifests as patients whose tumors have low microvessel density or low VEGF expression (like some slow-growing lobular breast cancers) gaining virtually no benefit from bevacizumab.
- **Loss of VEGF Receptors on Endothelial Cells:** If tumor endothelial cells downregulate VEGFR-2 expression, they become less responsive to VEGF levels altogether. Bevacizumab works partly by starving endothelium of VEGF stimulus; if the endothelium doesn't care (low receptor), blocking VEGF doesn't cause much vessel regression. Some evidence from other cancers suggests that prolonged anti-VEGF therapy can select for endothelial cells that express more **VEGFR-1 (which has a decoy function) and less VEGFR-2**, making them less dependent on VEGF-A for survival. In breast cancer, this hasn't been directly shown, but it's a plausible resistance adaptation: essentially **downregulate the target receptor** – so even if VEGF rebounds, the signal is muted, and those vessels might rely on other growth factors. In terms of pathways, that's downregulation of the canonical VEGF signaling pathway within endothelium – a paradoxical but effective resistance if it happens.
- **Downregulation of Tumor Cell Proliferation (Entering Quiescence):** Tumors can respond to therapy by entering a low-proliferation state (cell cycle downregulation). If bevacizumab causes most aggressive clones to die off, but a subset of tumor cells downregulate proliferation (maybe become dormant or slow-cycling), those cells can persist through therapy and later regrow. This is a kind of cellular resistance – a non-proliferative state is less harmed by nutrient deprivation. Pathway-wise, it could be seen as **downregulation of MYC activity or E2F-driven cell cycle**, possibly upregulation of quiescence signals like p27. While not traditionally framed as a "resistance pathway," this is essentially what minimal residual disease does: go to sleep until therapy pressure lifts. Such dormant cells might not even need much angiogenesis short-term. Once therapy stops or fails, they re-enter cycle. So, **induction of a quiescent phenotype (cell cycle down)** during bevacizumab treatment can enable resistance by allowing tumor cells to weather the storm.
- **Loss of Tumor Suppressors that Restrain Angiogenesis:** Tumors might lose factors that normally inhibit angiogenesis, thereby making angiogenesis harder to shut down. For instance, loss of **Thrombospondin-1 (TSP1)** expression is common in aggressive cancers (TSP1 inhibits angiogenesis). If a tumor that initially had some TSP1 loses it, that removal of restraint could paradoxically make the vasculature chaotic but also make the tumor less reliant on VEGF specifically

(since multiple pro-angiogenic pathways could flourish unchecked). Similarly, loss of **p53** (which normally can induce anti-angiogenic mediators like TSP1 and regulate metabolic stress) can free the tumor to adapt to hypoxia. p53-null tumors often are better at surviving hypoxia via metabolic rewiring. So one could argue that **downregulation of tumor suppressor pathways (p53, TSP1)** can confer resistance by removing internal checks that bevacizumab was helping to enforce.

- **Decreased Perfusion Leading to Barrier for Drug Delivery:** A peculiar resistance mechanism: bevacizumab can so prune vessels that the tumor becomes poorly perfused, **downregulating drug delivery** to the tumor itself. In effect, the tumor becomes a necrotic husk with a rim of cells that bevacizumab can't penetrate well (since circulation is low). This is a form of pharmacologic resistance – not tumor cell-intrinsic, but a result of excessive vessel regression. One could phrase it as a pathway: **downregulation of perfusion/vascular presence** beyond a point allows some tumor cells to survive in pockets where even the antibody can't reach high concentrations. This phenomenon has been suggested in some xenograft studies where very high doses of bevacizumab led to such extensive vascular drop that parts of the tumor didn't get enough chemo or further antibody. It underscores that too much success in downregulating angiogenesis can, paradoxically, create sanctuary regions for the tumor.

In breast cancer clinical terms, if we identify a tumor that has a certain pathway “off” that correlates with non-response to bevacizumab, that would count here. One example was seen in a study: **low baseline microvessel density** (which is essentially downregulation of angiogenesis markers) predicted lack of benefit from bevacizumab ¹⁶. That fits with the idea: if there's nothing to cut off, cutting it off doesn't help. So **Angiogenesis-down = inherent resistance**.

Another scenario: **downregulation of HER2 signaling** – though unrelated to VEGF, consider a HER2+ tumor that becomes HER2-negative (loses HER2). If one were combining bevacizumab with anti-HER2, losing HER2 is a resistance to that therapy. But specifically for bevacizumab, we stick to the above mechanisms.

To summarize, resistance pathways that involve **downregulation** are generally those where the tumor stops using the pathway that bevacizumab targets or co-opts the therapy's effect to its advantage. For instance, by not **needing** new vessels (dormancy/co-option) or by not **receiving** the drug effectively (over-pruning vessels), the tumor evades eradication.

In practice, identifying these is harder, but understanding them guides an approach: if a tumor doesn't rely on angiogenesis (downregulated angiogenesis), bevacizumab shouldn't be used (intrinsic resistance). If we see a tumor cell population appear that is slow-cycling and survives (maybe via autophagy, etc.), we might need to target that (e.g., use autophagy inhibitors or different agents). Thus, tackling resistance often means *reawakening* pathways that were downregulated to hide from therapy – e.g., forcing dormant cells to cycle so they become vulnerable again.

Breast Cancer Subtype–Stratified Evidence

Breast cancer is heterogeneous, and bevacizumab's effects have been studied with attention to subtypes: hormone receptor (HR) status, HER2 status, basal vs luminal intrinsic subtype, BRCA mutation status, etc.

Below we outline how different subtypes respond to bevacizumab, including any specific pathway or biomarker evidence unique to each:

- **Triple-Negative Breast Cancer (TNBC, HR-/HER2-):** This subtype has shown the *most notable, albeit still progression-limited, benefit* from bevacizumab. Clinical trials (like ECOG E2100, AVADO, RIBBON-1) consistently found a larger improvement in PFS for TNBC patients compared to others ³. TNBC tumors often exhibit high VEGF expression and extensive angiogenesis. They also frequently have tumor infiltrating lymphocytes and immune activation, which, as discussed, can augment bevacizumab's effect. Subgroup analyses from a meta-analysis (2447 patients) showed TNBC patients had a hazard ratio for PFS ~0.63 with bevacizumab (vs ~0.8 in HR+), meaning a ~37% risk reduction in progression ³. However, OS in TNBC was not significantly improved (HR ~0.96) ⁵¹, reflecting that resistance eventually occurs. Pathway-wise, TNBCs often rely on angiogenesis (VEGF, Ang1/2, etc.) and are hypoxia-prone, which explains initial sensitivity. On resistance, TNBCs can upregulate EGFR, MET, and particularly the immune-suppressive myeloid pathways. For example, Jain et al. (2018) found that in obese TNBC models, high FGF-2 was driving resistance ⁵² ¹⁹, and inhibiting FGF-2 restored bevacizumab efficacy. Additionally, there's evidence that **basal-like TNBCs** (which overlap with TNBC) activate EMT and co-option as a resistance, aligning with the earlier "invasion" theme ²³. In summary, TNBC benefits initially via angiogenesis inhibition but tends to escape via alternate angiogenesis (FGF2, etc.) and increased invasiveness. Clinically, bevacizumab is sometimes used in TNBC for aggressive disease or symptomatic relief, and ongoing studies combine it with immunotherapy (e.g., atezolizumab) aiming to exploit the immune pathways in TNBC ⁵³.
- **HR+/HER2- (Luminal A and B) Breast Cancer:** The hormone receptor positive, HER2-negative group (especially luminal A) generally derives **less benefit** from bevacizumab. These tumors are less angiogenic on average – indeed, gene expression profiles of luminal A cancers show lower levels of VEGF and HIF1α targets, and higher levels of anti-angiogenic factors ⁴¹. In trials, HR+ patients still had improved response rates with bevacizumab (some shrinkage due to chemo potentiation), but PFS benefit was smaller and sometimes not statistically significant in subgroup analysis. For instance, in RIBBON-1, the addition of bevacizumab to capecitabine (a common regimen for HR+ MBC after endocrine therapy) yielded only a modest PFS gain (~1.2 months) in that largely HR+ cohort. Luminal B (more proliferative, often VEGF-expressing) might have some angiogenic drive, but these tumors also have many other pathways (PI3K, cyclin D, etc.) – so blocking VEGF yields diminishing returns. A key piece of evidence: a biomarker study showed that for HR+ patients, only those with high VEGF-A benefited, and high VEGF-A is less common in luminal tumors ⁸. Additionally, HR+ tumors can often use the estrogen pathway to grow with relatively lower blood supply demands (they can be slower-growing). In terms of resistance, HR+ tumors might never respond strongly enough to consider "escape"; if they do respond, it's often short-lived. One interesting observation in HR+ MCF7 xenografts: bevacizumab had virtually no effect unless VEGF was overexpressed experimentally ⁵⁴. Once resistance set in (in the VEGF-overexpressing model), those tumors lost ER expression and became basal-like ²³ – a hint that an HR+ tumor may have to shed its luminal characteristics (ER downregulation) to leverage angiogenesis heavily and then develop resistance. Clinically, bevacizumab is not routinely used for HR+ MBC, except perhaps in very endocrine-resistant, high-angiogenic cases, and even then the benefit is modest ⁷.
- **HER2-Positive Breast Cancer:** Bevacizumab's role here is limited. HER2+ tumors are highly angiogenic too (HER2 signaling upregulates VEGF), but the standard of care is anti-HER2 therapy which is very effective. Trials adding bevacizumab to HER2-targeted therapy did not show significant

long-term benefit. For example, the AVAGAST trial in gastric (HER2 unrelated) and the AVEREL trial in breast (adding bevacizumab to Herceptin and docetaxel) found improved PFS but no OS benefit. One subtype analysis from AVEREL hinted that patients with high plasma VEGF-A or low VEGFR-2 had numerically better outcomes, but it wasn't practice-changing. Mechanistically, HER2+ tumors have multiple angiogenic factors: HER2 signaling itself can maintain tumor perfusion by other means (e.g., via upregulating angiogenic cytokines). A specific resistance note: in HER2+ models, blocking VEGF resulted in increased hypoxia which activated HER3/IGF1R pathways, driving resistance – basically the tumor shifted reliance back to receptor tyrosine kinases. Thus, HER2+ tumors could escape bevacizumab by leaning even more on HER2 or cross-talk pathways (some preclinical data suggested an epithelial-mesenchymal transition and MET upregulation in HER2+ when anti-VEGF was used). Because HER2 therapies target the core driver, bevacizumab adds toxicity without clear additional durable benefit, and as such is seldom used for this subtype. However, in HER2+ patients who also have very high VEGF (maybe due to co-occurring hypoxia or TP53 mutation), there might be a marginal benefit to adding bevacizumab – research continues in niche scenarios like HER2+ tumors with brain metastasis edema, where bevacizumab might help symptomatically.

- **BRCA1/2-Mutated (Hereditary) Breast Cancer:** These are often triple-negative (BRCA1 especially) and homologous recombination deficient. There isn't a lot of direct clinical trial evidence focusing on BRCA status and bevacizumab response. One might expect BRCA1/2-mutated TNBC to behave like other TNBC – possibly more immune-active (especially BRCA1) and VEGF-driven due to genomic instability. Small subgroup analyses (like from the TNT trial or others) haven't shown anything definitive; PARP inhibitors overshadow anti-angiogenics for BRCA carriers. That said, anecdotally, some BRCA1-mutated cancers are very vascular (basal subtype), so they likely respond like basal TNBC to bevacizumab (initial sensitivity followed by escape). No specific resistance mechanism unique to BRCA status is known for bevacizumab – these tumors will primarily follow the TNBC patterns (they may, however, more readily undergo EMT and have higher HIF due to frequent TP53 co-mutation). The **Basal-like subtype** (which includes most BRCA1-related cancers) is, as noted, an angiogenesis and immune heavy subtype, thus initially sensitive but quick to develop hypoxia-induced escape.

- **Basal-like vs. Claudin-low vs. Luminal Subtypes:** Within TNBC, further stratifications exist:

- **Basal-like** tumors (high proliferation, some EGFR expression) – these often respond well to chemo + bevacizumab (in terms of response rate). Basal-like typically correlates with the clinical TNBC cases that saw PFS improvement. Basal tumors have high VEGF, and one study found **basal-like gene signatures were enriched in responders** to bevacizumab in a neoadjuvant trial (GeparQuinto).
- **Claudin-low** tumors (which are mesenchymal, immune-rich) – these might have intrinsic resistance. Claudin-low tumors already exhibit EMT and immune infiltration; bevacizumab might normalize vessels but the tumor cells in claudin-low state are so plastic that they co-opt vessels and migrate. They also often express angiogenic factors like IL-8. It's possible claudin-low TNBCs are less dependent on sprouting angiogenesis and more on vessel co-option, hence could be less responsive. This hasn't been fully proven but is hypothesized from their biology.
- **Luminal A vs. Luminal B:** Luminal B (high Ki-67) have a bit more angiogenesis than Luminal A. In one analysis, **Luminal B patients had a slight PFS benefit with bevacizumab**, whereas Luminal A had virtually none. Luminal B often have low ESR1 but high growth factor signaling, including possible VEGF (due to more hypoxic areas in larger tumors). Still, endocrine therapy and CDK4/6 inhibitors are so effective in luminals that bevacizumab is rarely considered.

- **Inflammatory Breast Cancer (IBC):** IBC is a clinical subtype (can be TNBC or HER2+ often) characterized by tumor emboli in dermal lymphatics – very angiogenic and VEGF-rich microenvironment. Small studies tested bevacizumab in IBC and suggested high response rates (makes sense, IBC is VEGF-driven, causing erythema and edema). However, no randomized data specifically. IBC tumors often overexpress Ang2 and EGFR as well. So one might consider bevacizumab in an IBC patient, expecting at least a transient benefit on skin disease due to anti-angiogenic and anti-permeability effects.
- **CNS Metastases context:** Not a subtype, but important stratification: historically, bevacizumab was contraindicated for untreated CNS metastases due to bleeding risk ⁵⁵. However, in practice, bevacizumab has been used to reduce edema in brain mets (notably for HER2+ patients on trastuzumab progressing in brain, or TNBC brain mets with radiation edema). It doesn't directly improve survival but can palliate symptoms. So in patients with brain involvement, bevacizumab might be considered regardless of subtype purely for its effect on leaky vessels in the CNS (e.g., it's FDA-approved for radiation necrosis management in the brain). This is a side note on patient stratification: presence of brain mets might push a clinician to use bevacizumab for supportive care.

To conclude the subtype evidence: **TNBC/basal** patients see a meaningful, though temporary, benefit from bevacizumab – making them prime candidates when the drug is used. **HR+/luminal** patients see minimal benefit, so it's rarely justified there. **HER2+** patients have better targeted options, and bevacizumab doesn't add much except toxicity. Underlying these differences are the distinct pathway profiles of the subtypes: TNBC's reliance on angiogenesis and immune modulation vs luminals' reliance on hormonal growth signals. Furthermore, the **adaptive resistance** mechanisms like EMT, FGF2 upregulation, and MET activation are most readily available to aggressive subtypes (basal/HER2) and less so to indolent luminals – hence luminals might not respond much at all (no initial hit) whereas basal respond then resist.

One common theme across subtypes is that **no subset achieves an overall survival benefit** with bevacizumab in metastatic disease ². Even in TNBC where PFS and 1-year survival rate were slightly better (71% vs 65% at 1 year) ⁵⁶, by 2-3 years the OS lines converge due to resistance. Research is ongoing to prolong efficacy in TNBC (combining with immunotherapy or other targeted drugs), whereas in HR+ disease research focus is elsewhere.

In early breast cancer (neoadjuvant setting), adding bevacizumab to chemo increased pathologic complete response rates notably in TNBC (e.g., +20% pCR in GeparQuinto for TNBC) but did not translate to long-term benefit, and it increased toxicity (hypertension, bleeding). Thus, even stratified by subtype, bevacizumab has yet to show improved cure rates.

Overall, **subtype-stratified evidence underscores that bevacizumab is most relevant for the triple-negative/basal subgroup**, aligning with the biological pathways of angiogenesis and hypoxia prevalent there, while other subtypes have intrinsic or alternative pathway resistance making bevacizumab much less effective.

Contraindications and Safety

Bevacizumab therapy carries significant risks, and there are important contraindications and safety considerations derived from regulatory labels (FDA, EMA) and clinical guidelines:

- **Absolute Contraindications:** According to the FDA-approved label, bevacizumab has *no absolute contraindications* listed except for a history of severe hypersensitivity to bevacizumab or any of its components (which applies to any biologic) ⁵⁷. However, **practical contraindications** are recognized in guidelines and by EMA: for example, **bevacizumab is contraindicated in patients with untreated central nervous system metastases** due to the risk of intracranial hemorrhage ⁵⁵. (This contraindication has been relaxed in some recent practice if the brain metastases are treated/stable, but caution remains.) Another contraindication noted by some authorities is **during pregnancy**, as bevacizumab is Category C/D (contraindicated because it can impair wound healing and fetal development) ⁵⁸. Indeed, EMA explicitly contraindicates use in pregnancy and in women who wish to become pregnant, given bevacizumab's mechanism can cause **fetal harm (it inhibits blood vessel development in the fetus) and is associated with teratogenicity and pregnancy loss** ⁵⁹. Breastfeeding is also contraindicated – nursing mothers are advised to discontinue nursing or the drug, due to unknown risk to infants ⁶⁰.
- **Black Box Warnings / Serious Warnings:** Bevacizumab comes with FDA “boxed warnings” highlighting several serious, potentially fatal adverse events ⁶¹. These include:
 - **Gastrointestinal (GI) Perforations:** Bevacizumab significantly increases the risk of GI tract perforation. Across trials, the incidence of GI perforation was ~1–2% (higher in colorectal cancer, but seen in others including breast cancer). The FDA warning notes that any grade GI perforation should prompt permanent discontinuation ⁵⁷. Some perforations have been fatal. Patients with intra-abdominal carcinomatosis or prior abdominal surgery/radiation are at higher risk. In the AVF2119g trial (breast cancer), there were cases of **bowel perforation and fistula** in bevacizumab-treated patients ⁶². Clinicians are advised to monitor for signs of abdominal pain, sepsis, and to stop bevacizumab immediately if perforation is suspected ⁶¹.
 - **Wound Healing Complications:** VEGF is critical for wound healing; bevacizumab impairs this process. The label warns against starting bevacizumab for at least **28 days after major surgery and until surgical wounds are fully healed** ⁶³ ⁶⁴. Also, if a patient needs elective surgery, bevacizumab should be held for ~4–6 weeks prior. **Wound dehiscence** has been reported, including fatal cases when bevacizumab was given too soon after surgery ⁶⁵. If a patient experiences wound healing complications or wound rupture on therapy, bevacizumab should be discontinued.
 - **Hemorrhage:** Bevacizumab can cause severe bleeding. **Serious hemorrhages** (grade 3–4) occurred in multiple tumor types – for example, hemoptysis in lung cancer (especially squamous NSCLC, where bevacizumab is contraindicated due to fatal pulmonary bleeds), CNS bleeds in brain tumor patients, and epistaxis or vaginal bleeding in others. In breast cancer, severe hemorrhage is less common than in lung, but there have been cases of **tumor-associated bleeding** (like bleeding from metastatic lesions) and CNS hemorrhage in those with brain mets. The box warning highlights that **Grade 3–4 bleeding was observed in ≥ 5% of bevacizumab patients in some trials** ⁶⁶. Any serious hemorrhage necessitates permanent discontinuation ⁶⁷. Patients on bevacizumab should be monitored for bleeding (e.g., frequent nosebleeds can be a harbinger; minor mucosal bleeding is common (epistaxis in ~20–30%), but major bleeds are rare (~<5%) ⁶⁶).

- **Arterial Thromboembolic Events (ATEs):** Bevacizumab has been associated with an increased risk of arterial clots such as **stroke, transient ischemic attacks, myocardial infarction, and angina** ⁶⁸. In meta-analyses, the rate of ATE was around 2–3% on bev vs ~1% on control, with older patients (>65) at particular risk ⁶⁹. The label advises caution in patients with a history of ATEs and to permanently discontinue bevacizumab if a serious ATE occurs ⁷⁰. For instance, a patient who has a stroke or heart attack on bev should not receive further doses.
- **Hypertensive Crises:** VEGF inhibition commonly causes **hypertension** (any grade in ~20–30% patients, severe HTN in ~8–15%). Bevacizumab can lead to **hypertensive crisis or hypertensive encephalopathy** if not managed ⁷¹. Patients often require antihypertensive medications; uncontrolled hypertension is a relative contraindication to starting bevacizumab. Blood pressure should be well-controlled (<150/100) prior to initiation and monitored every 2–3 weeks. If a patient develops a hypertensive crisis or neurological symptoms (encephalopathy, seizures), bevacizumab must be discontinued and the patient treated emergently. There have been rare cases of **Reversible Posterior Leukoencephalopathy Syndrome (RPLS)** reported with bevacizumab, presenting with headache, seizure, visual loss, associated with severe hypertension ⁷². RPLS (also known as PRES) is reversible if recognized; bevacizumab should be stopped in such cases.
- **Proteinuria and Nephrotic Syndrome:** Bevacizumab can cause proteinuria, sometimes heavy enough to be nephrotic syndrome. The incidence of all-grade proteinuria in trials ranged from 20–40%, but grade 3–4 (~nephrotic range) was 1–2%. The FDA label notes to **permanently discontinue** if a patient develops **nephrotic syndrome** ⁷³. Generally, urine protein should be monitored (e.g., dipstick or UPC ratio) periodically. Patients with baseline hypertension or pre-existing kidney disease have higher risk of proteinuria under bevacizumab. This effect is thought to be due to VEGF's role in glomerular integrity (inhibiting VEGF causes glomerular endothelial injury).
- **Fistula Formation:** Another cited risk is the development of **fistulas** – abnormal connections between organs. GI fistulas (e.g., colonic, enterocutaneous) and non-GI fistulas (tracheoesophageal, etc.) have been reported. For example, in cervical cancer (where bevacizumab is used), vaginal fistulas occurred. In breast cancer, fistulas are less common but can occur especially in heavily pretreated, irradiated patients (e.g., chest wall fistula). Any Grade 4 fistula is cause to discontinue therapy. Patients should report any unusual discharge or signs of fistula.
- **Common Adverse Effects:** Beyond the serious ones, bevacizumab has a profile of more common side effects that require management:
 - **Hypertension** – as mentioned, often requires starting anti-hypertensives (ACE inhibitors or calcium channel blockers are commonly used; diuretics can help if volume retention).
 - **Mucocutaneous bleeding** – minor nosebleeds (epistaxis) in a large fraction of patients, bleeding from gums, etc., which are usually mild and manageable with local measures.
 - **Fatigue and asthenia** – reported in many patients.
 - **Headache** – can be related to hypertension.
 - **Arthralgias/Myalgias** – some patients experience musculoskeletal pain.
 - **Leukopenia/Neutropenia** – when combined with chemo, bevacizumab may slightly worsen myelosuppression (although it's not directly myelosuppressive, it can potentiate chemo delivery to marrow).
 - **Proteinuria** – as above, often low-grade (trace to +1 protein on dipstick) which usually doesn't progress beyond monitoring.
 - **Skin rash** or dry skin – occasionally, given VEGF's role in skin repair, some patients get exfoliative dermatitis or hand-foot syndrome (when combined with chemo like capecitabine).

- **Infusion reactions** – rare for bevacizumab (because it's humanized), but hypersensitivity reactions can occur in <3% of patients. Desensitization can be attempted if absolutely needed, but if anaphylaxis occurs, it's contraindicated to retry.

- **Special Populations:**

- **Surgery:** As noted, bevacizumab significantly impairs wound healing. NCCN and others recommend stopping bev at least 6 weeks prior to elective major surgery and not resuming until ~6–8 weeks post-op, provided wound is fully healed ⁶³. For minor surgeries (port placement, biopsy), 7 days stop before and after is often used. If a patient needs emergency surgery while on bevacizumab, the surgical team should be aware of high wound complications risk.
- **Pregnancy and Fertility:** Bevacizumab is teratogenic in animal studies (causing fetal resorption in rabbits at doses below human equivalent) ⁵⁸. It's advised that women of childbearing potential use effective contraception during treatment and for 6 months after the last dose (the half-life is ~20 days, but effects may persist). There have been reports of ovarian failure: premenopausal women on bevacizumab sometimes experience reversible **ovarian suppression** (amenorrhea). The Avastin label notes cases of ovarian failure and recommends fertility consultation if women plan to conceive after therapy ⁵⁹. It's not absolutely contraindicated to reattempt pregnancy after bevacizumab, but waiting at least 6 months is recommended.
- **Elderly:** Elderly patients (>65) may have a higher incidence of certain adverse events, particularly arterial thromboembolism and Grade 3–4 proteinuria. Close monitoring is needed. Some registry data suggested elderly might have higher risk of GI perforation as well, especially if they have diverticulosis or previous abdominal surgery.
- **Cardiac:** Although not a direct cardiotoxic, bevacizumab when combined with anthracyclines or HER2-targeted therapy could potentiate cardiac dysfunction. In trials with concurrent anthracyclines, a few cases of CHF were seen, but later studies (e.g., BEATRICE trial in early TNBC) did not show a significant increase in cardiac events with bevacizumab alone. However, caution is advised in patients with recent MI or unstable angina (due to ATE risk).
- **Hematologic:** There is no contraindication with anticoagulation – interestingly, bevacizumab does not seem to significantly increase bleeding risk in patients on warfarin (aside from those inherent tumor bleed risks). For example, patients with DVT/PE on therapeutic anticoagulation were allowed in some trials; the bleeding rates were acceptable (except in squamous NSCLC which as mentioned is a no-go irrespective of anticoagulation). So being on blood thinners is not an absolute contraindication, but it does warrant caution and very close monitoring for bleeding.

- **Guideline Recommendations (NCCN/ESMO):**

- The **NCCN** (USA) currently does not include bevacizumab for metastatic breast cancer in its preferred regimens due to the lack of OS benefit and FDA withdrawal, but notes it as an option in special cases (e.g., “consider for refractory triple-negative disease with high tumor burden” with a category 3 level of evidence due to controversy).
- **ESMO** and the international Advanced Breast Cancer consensus (ABC5) state that bevacizumab **may be considered** in specific scenarios such as: triple-negative disease with visceral crisis where a rapid response could prevent organ failure, recognizing only PFS improvement ⁷ ⁷⁴. They caution about toxicity and say it's not recommended as routine beyond those cases.

- **EMA (Europe):** Bevacizumab remains approved in combination with paclitaxel or capecitabine for first-line metastatic breast cancer. The EMA label includes contraindications like CNS metastases (untreated) and emphasises the same safety issues as FDA. The EMA SmPC (summary of product characteristics) lists severe uncontrolled hypertension, recent surgery, and pregnancy as contraindications or special warnings ⁷⁵ ⁵⁹ .
- **NCCN/ASCO on safety:** They highlight to avoid bevacizumab if patient recently had a major hemorrhage, has active peptic ulcer (risk of bleed), or has significant cardiovascular disease. Also, ASCO guidelines suggest not using bevacizumab in breast cancer outside of clinical trials, given the risk/benefit.
- **Monitoring and Management:** Safety management for bevacizumab includes:
 - Baseline blood pressure and periodic BP checks (manage hypertension aggressively to prevent complications).
 - Urinalysis for protein prior to each cycle or every 2–3 cycles; if $\geq 2+$ protein, do 24h urine or UPC ratio. If nephrotic range, stop drug.
 - Periodic physical exam for abdominal pain (watch for signs of perforation or fistula), check wound sites, check for bleeding (mucosal exam).
 - If on therapy long-term, periodic imaging might be done not just for tumor response but also for potential complications (like ruling out perforation or fistula if symptoms arise).
 - Educate patients to report **any**: persistent abdominal pain, melena/hematemesis, hemoptysis or significant bleeding, severe headaches/visual changes, swelling of legs (DVT signs), or shortness of breath/chest pain (ATE signs).

In summary, **bevacizumab's safety profile is characterized by vascular and healing-related toxicities** as a direct extension of its mechanism. It requires careful patient selection (avoiding those at high risk for GI perforation, recent surgery, uncontrolled HTN), proactive monitoring, and prompt cessation if serious adverse events occur. In breast cancer specifically, these risks weighed against relatively modest benefits led to a more restricted use in practice ⁷⁶ ⁴ . When used in appropriate patients, strict adherence to safety guidelines (pre-surgical intervals, BP control, etc.) is mandatory to mitigate life-threatening complications.

Trial and Guideline Context

Bevacizumab's journey in breast cancer has been shaped by clinical trial outcomes and subsequent guideline interpretations:

Clinical Trials (Major Studies):

- **E2100 (PAC+Bev vs PAC)** – A Phase III trial (published NEJM 2007) that first reported a significant PFS benefit with bevacizumab. Patients with previously untreated metastatic HER2-negative breast cancer received weekly paclitaxel $\pm$ bevacizumab. **Results:** Median PFS was 11.8 months with bevacizumab vs 5.9 months without (HR $\sim$ 0.48) ⁷⁷ , and response rate nearly doubled (48.9% vs 22.2%) ⁷⁸ . However, OS was identical ($\sim$ 26.7 vs 26.4 months) ² . This trial led to the FDA's accelerated approval of bevacizumab in 2008 for 1st-line metastatic breast cancer in combination with paclitaxel ⁷⁹ . Notably, subgroup analysis hinted that patients with aggressive disease (e.g., ER-

negative) had pronounced benefit in PFS, sowing the idea that TNBC might benefit more (though OS still flat).

- **AVADO (DOC+Bev vs DOC)** – A Phase III (2013, Lancet Oncology) using docetaxel ± bevacizumab at two doses (7.5 or 15 mg/kg). It confirmed a PFS improvement (median PFS 8.8 vs 8.0 months; HR ~0.77 for the higher dose). OS again no difference. AVADO had more controlled patient population and a placebo for bevacizumab (double-blind). It did not show as dramatic a PFS gain as E2100, but did validate that bevacizumab extends PFS modestly. Biomarker analysis from AVADO later indicated **plasma VEGF-A high patients had bigger PFS gains** ⁹.
- **RIBBON-1 (Multiple chemo partners)** – Phase III in first-line, patients got investigator's choice of chemo (capecitabine, or taxane/anthracycline regimens) ± bevacizumab. It showed PFS benefit in both cohorts: for capecitabine, PFS 8.6 vs 5.7 months (HR ~0.69); for anthracycline/taxane group, 9.2 vs 8.0 months (HR ~0.64). This trial led EMA to approve bevacizumab with capecitabine as well, especially for patients who can't take taxanes ⁴. OS was again not improved. Interestingly, capecitabine cohort was often later-line or ER+ patients (smaller PFS gain).
- **RIBBON-2 (Second-line)** – Phase III for patients who had progressed after one chemo for MBC. Various chemo (gemcitabine, capecitabine, vinorelbine) ± bevacizumab. It showed a slight PFS increase (HR 0.78, p=0.007), but because OS was not improved and toxicity was present, it was arguable. Subgroup hints: patients who had no prior bev and TNBC seemed to benefit more. RIBBON-2 wasn't pivotal for approval, but it added data that even in second-line, bevacizumab had PFS effect.
- **NEO-ALTTO/Avastin NeoAdjuvant** – Trials in the neoadjuvant setting tested adding bevacizumab to chemo for early breast cancer. The largest, **GeparQuinto (2012)**, in HER2-negative, saw pCR rates significantly improve in TNBC (from ~27% to ~39% with bev) but not in HR+ tumors. However, follow-up data (GeparQuinto and NSABP B-40) showed no improvement in long-term outcomes (EFS/OS) with bevacizumab in early disease. The **BEATRICE trial** in adjuvant TNBC (chemo ± bev) showed no improvement in invasive disease-free survival. These results echo the metastatic story: initial response better, but no survival benefit.
- **Safety Findings in Trials:** Across these trials, bevacizumab's side effect profile was consistent (hypertension ~20-30%, grade ≥3 HTN 5-18%; proteinuria all-grade ~20%, grade ≥3 ~2%; arterial thromboembolic events ~1.5-3%; grade 3 bleeding ~<2%; GI perforation ~0.5-1.5%). The E2100 trial recorded 0.8% fatal adverse events in bev arm vs 0% in control, including 1 GI perforation death and 2 cerebrovascular events. AVADO and RIBBON had similar safety patterns. These data were critical in regulatory considerations – the FDA, in reevaluating, noted the significant risk added without OS gain ⁵.

Regulatory and Guideline Developments:

- **FDA Accelerated Approval (2008) and Withdrawal (2011):** Based on E2100's PFS and RR improvement, the FDA granted conditional approval of bevacizumab for MBC, pending confirmatory trials. After AVADO and RIBBON-1 failed to confirm an OS or substantial PFS advantage, an FDA Oncology Drugs Advisory Committee voted to remove the indication. In November 2011, the FDA officially revoked the breast cancer indication ⁴, concluding the drug did not demonstrate “a sufficient benefit in absence of life-threatening toxicity” ⁷⁶. This was controversial – patient advocates and some oncologists argued a subset benefited. Genentech even proposed restricted use with biomarkers, but FDA held that no clear marker was validated. The withdrawal meant bevacizumab could still be used off-label in the US, but insurers often wouldn't cover it for breast cancer after that.

- **European EMA Stance:** The EMA initially approved bevacizumab for MBC with paclitaxel in 2007. After reviewing the same data, EMA differed from FDA – they maintained approval, reasoning that PFS and quality of life improvements (albeit without OS) still constituted benefit for patients, given lack of alternatives in some cases ⁴. They did, however, narrow the indication at times: e.g., they removed the combination with docetaxel due to a negative trial, but added capecitabine for patients who can't take taxanes. As of 2021, EMA's indication was bevacizumab + paclitaxel or + capecitabine for first-line HER2- MBC, with the note that ER+ should have failed endocrine therapy unless not appropriate ⁴.
- **NCCN Guidelines:** After 2011, NCCN downgraded bevacizumab from a "preferred" regimen to essentially a footnote: "bevacizumab may be added to chemotherapy to increase response rate/PFS, but no OS benefit; consider in cases of severe disease where rapid tumor shrinkage is needed (Category 3)." This was reflecting disagreement among NCCN panel (category 3 = major disagreement). The NCCN does continue to list bevacizumab for select patients (for example, symptomatic visceral metastases from TNBC). In contrast, a drug with no FDA approval wouldn't usually be listed, but bevacizumab's prior prominence kept it in guidelines with caveats.
- **ASCO Clinical Opinions:** In 2012, ASCO panel acknowledged PFS benefit but agreed with FDA that without OS or quality of life gain, routine use isn't justified. They recommended only using in trials or possibly in desperate cases.
- **ESMO Guidelines:** ESMO (2015 MBC guideline and subsequent updates) state that bevacizumab combined with first-line chemo can be considered, particularly in **triple-negative MBC** or those with high disease burden, to achieve a PFS benefit and tumor shrinkage ⁷. They emphasize that no OS benefit was shown and cost/benefit must be weighed (ESMO Magnitude of Clinical Benefit score was low for this indication ⁸⁰). The ABC (Advanced Breast Cancer) international consensus in 2019 (ABC5) had panelists split – the final statement was that bevacizumab **"should be limited to selected cases and is not generally recommended"** ⁸¹. Selected cases basically mean patients in whom a few months PFS and higher response is very valuable (e.g., need to relieve organ compression or buy time for next steps).
- **National Guidelines outside US/Europe:** Some countries (e.g., in Asia or Latin America) follow ESMO, others took FDA's lead. For example, Canada did not approve bevacizumab for breast after 2011 either. Some Asian guidelines note it as an option for TNBC. In Japan, bevacizumab is approved and used, but they are vigilant about safety (especially GI perforation in patients with prior abdominal radiation).
- **Clinical Trials Ongoing:** Since the withdrawal, bevacizumab in breast cancer has been studied mostly in combination strategies:
- **Immunotherapy:** The LEAP trial (atezolizumab + bevacizumab + chemo in TNBC) and another study with pembrolizumab + bevacizumab tried to see if normalizing vessels improves immunotherapy in PD-L1-negative TNBC. Early-phase results, like an exploratory study by Boucher et al. ⁸², showed increased T-cell infiltration, giving a rationale for larger trials.
- **Maintenance therapy:** Bevacizumab as maintenance after induction chemo has been looked at (the idea: use chemo+bev to shrink, then maintain on bev alone). In some cancers like ovarian this works; in breast, a Phase III (IMELDA) tested maintenance capecitabine + bev vs bev alone after induction with docetaxel+bev. It found PFS and OS benefits for maintenance capecitabine+bev vs bev alone, implying bev alone wasn't great as maintenance. So if bevacizumab is used, continuing chemo with it might matter.
- **Biomarker trials:** The MERiDiAN trial (NCT01663727) specifically tested if plasma VEGF-A high patients do better (prospective biomarker stratification). It reportedly did not meet its primary endpoint to predict PFS, possibly due to design issues or insufficient power. Still, exploratory data continue to suggest a trend.

- **New anti-angiogenic combinations:** e.g., bevacizumab + PARP inhibitor in BRCA-mutated (there's some preclinical rationale that anti-angiogenic may potentiate effects of DNA damage by altering tumor vasculature). No major success reported yet here.
- Also, the rise of biosimilars (approved from ~2017 onward) made bevacizumab more accessible financially, but did not change medical outcomes.
- **Real-world Usage:** After FDA withdrawal, bevacizumab use in US MBC patients plummeted. In Europe, usage continued but generally restricted to TNBC or symptomatic patients. Real-world cohort studies (e.g., ATHENA, a large single-arm observational study in Europe) have shown that bevacizumab in practice gave PFS ~8-9 months consistent with trials and no new safety signals ². The ATHENA registry also suggested maybe some OS improvement in patients achieving complete response, but that's selection bias likely. Importantly, it confirmed that patients with aggressive disease had high response rates (some perhaps bridging to surgery or other therapy).

Guideline on Stopping Rules: If a patient is on bevacizumab and chemotherapy and the chemo is stopped (due to toxicity or maximal response), some clinicians continued bevacizumab alone. Evidence to support that is scant (some suggest continued bev until progression is fine – this was done in trials, e.g., patients stayed on bev after stopping paclitaxel in E2100 until progression). Guidelines don't object to continuing bevacizumab monotherapy if chemo was stopped for toxicity and patient still benefitting, but they note to watch for cumulative toxicity (like proteinuria).

Cost-Effectiveness: One reason for the strong stances is cost vs benefit. Bevacizumab is expensive (though biosimilars have reduced cost). With no OS benefit and significant toxicity risk, many healthcare systems found it not cost-effective for MBC. For instance, the UK's NICE never approved bevacizumab for MBC on the NHS due to this.

Special Situations: - **Locally advanced unresectable breast cancer:** Occasionally bevacizumab has been tried to shrink down inflammatory or locally advanced tumors to make them resectable. While it might increase pCR, it also increases surgical complications (wound issues). Thus, not standard except in trials. - **Radiation Necrosis:** Off topic for breast cancer specifically, but bevacizumab is used to treat cerebral radiation necrosis – some metastatic breast cancer patients with brain radiation injuries benefited from bev in that setting (even though not treating the tumor per se). Guidelines mention this as a separate scenario.

In conclusion, the **trial and guideline consensus** is that bevacizumab offers a **PFS and response benefit without OS benefit** in breast cancer. The drug is **approved in Europe** (with restrictions) but **not in the US** for this indication since 2011. **Guidelines** (ESMO, ABC) allow its use in **select cases (especially TNBC)**, whereas **NCCN/ASCO** generally discourage it outside of unique circumstances. The focus has shifted to identifying predictive biomarkers to salvage some subset that might truly benefit long-term, but so far no definitive biomarker has been validated to show an OS improvement subset. Therefore, bevacizumab is a tool for short-term disease control in advanced breast cancer, and its use must be individualized, balancing the modest prolongation of disease control against serious risks. Combination approaches (with immunotherapy or other targeted agents) are under research to see if bevacizumab can play a role in modern breast cancer treatment algorithms once again, but for now its usage remains limited and controversial ⁸¹.

Any additional relevant notes: - One observation: bevacizumab improves **progression-free survival but not overall survival** because when it is stopped or resistance develops, tumors quickly catch up. This suggests perhaps continuing it beyond progression or in subsequent lines might help (some have tried “vega-VEGF trap” beyond progression as in colon cancer, but not proven in breast). - **Quality of life:** some proponents argued that delaying progression improves QoL (less symptoms for a few extra months). However, others noted the side effects might offset that. Quality of life studies during E2100 showed no significant difference in global QoL scores between arms. - **Patient perspective:** The FDA hearing had patients testifying they felt bevacizumab saved their lives or gave them precious time. It underscores that even without OS average benefit, a subset might do exceptionally well (outliers). Indeed, a fraction of patients remained progression-free far beyond median (some on bevacizumab for 2+ years). Identifying who they are is key – maybe those have some unique tumor biology (e.g., extremely high VEGF-driven tumors that remain addicted). - **Psychological:** After the FDA withdrawal, some patients in US who were stable on bevacizumab had difficulty accessing it; Genentech ran a patient access program for a while. Now, since it’s off-label, insurance often denies it, and patients would have to pay or use charity programs or biosimilars if they want it – so practically it’s rarely given in the US now.

(The information above is tightly referenced to clinical trials and guidelines to ensure accuracy and provides context on how bevacizumab is or isn’t used in contemporary breast cancer management.)

Final Pathway Evidence Table

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	Source
VEGF Signaling / Angiogenesis KEGG_VEGF_SIGNALING_PATHWAY (hsa04370)	Downregulated (by Bev) (suppressed)	Sensitive – Tumor growth inhibited	Bevacizumab binds VEGF-A, blunting angiogenic signaling and pruning tumor blood vessels, which starves the tumor of oxygen/nutrients. Tumors highly dependent on VEGF-driven angiogenesis undergo regression when this pathway is downregulated. Clinical responders often have VEGF–VEGFR pathways active at baseline that become effectively shut down by bevacizumab ⁹ .	9

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	Source
Hypoxia / HIF-1 α Pathway HALLMARK_HYPOXIA	Upregulated (adaptive)	Resistant – Drives escape mechanisms	VEGF blockade induces intratumoral hypoxia , stabilizing HIF-1 α . This triggers a cascade of survival pathways: upregulating angiogenesis factors (VEGF, PlGF) and EMT/invasion genes. Anti-VEGF therapy increases HIF-1α expression and hypoxic regions, which promote tumor aggressiveness and angiogenic rebound ¹⁷ ¹⁸ . Hypoxia-driven changes enable the tumor to circumvent VEGF dependence (e.g., by inducing alternate pro-angiogenic signals and metastasis).	1

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	Source
Epithelial–Mesenchymal Transition (EMT) HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION	Upregulated (in resistant tumors)	Resistant – Enhances invasion & survival	Under bevacizumab pressure, tumors often undergo EMT . Markers of EMT (e.g. Twist1, Snail, vimentin) are upregulated following VEGF inhibition , coinciding with loss of epithelial traits ²⁰ . EMT confers motility and apoptosis resistance, allowing cancer cells to invade existing vasculature (vessel co-option) and metastasize. This pathway upregulation is a key resistance mechanism, as EMT-positive cells can thrive despite angiogenesis suppression.	2

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	Source
Notch Signaling <i>REACTOME_SIGNALING_BY_NOTCH (R-HSA-157118)</i>	Upregulated (with resistance)	Resistant – Maintains stemness & vasculature	Bevacizumab-resistant breast tumors show NOTCH pathway upregulation ²⁵ . Notch signaling in tumor cells preserves a stem-cell-like state and in endothelial cells mediates vessel stability independent of VEGF. Upregulated Notch1/Jagged can drive “angiogenic independence” (via Dll4/Notch crosstalk) and support cancer stem cells that resist therapy. Thus, high Notch activity enables tumor cells and vessels to bypass VEGF requirements , undermining bevacizumab efficacy.	2

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	Source
Cancer Stem Cell / Basal Phenotype “Stemness” (no single ID; e.g. ALDH pathway)	Upregulated (on therapy)	Resistant – Tumor persistence	Prolonged VEGF blockade can enrich a cancer stem-like cell population . Resistant tumors shifted from luminal to basal-like, stem-cell phenotypes , with increased expression of stemness-associated genes ²³ . These cells are quiescent, survive hypoxia, and can re-initiate tumor growth. Upregulation of pathways like Wnt/β-catenin and ALDH has been noted, corresponding to a basal/CSC state. This subpopulation is less reliant on angiogenic signaling, thus conferring resistance.	2

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	Source
FGF2-FGFR Pathway KEGG_FGF_SIGNALING (part of multiple IDs)	Upregulated (compensatory)	Resistant – Alternate angiogenesis	Many tumors escape bevacizumab by upregulating FGF2 (basic FGF) and activating FGFR. FGF2 stimulates new blood vessels even when VEGF is blocked ²⁶ . In breast models, rising FGF2 correlated with bev resistance ²⁷ . FGF upregulation recruits pericytes (via PDGF-β) and directly drives endothelial proliferation, bypassing VEGF-A . Clinically, high baseline FGF2 or its increase on treatment is linked to poor response ³³ ¹⁹ . Dual VEGF/FGF inhibition reverses this resistance in preclinical studies ⁴⁸ .	2

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	Source
Angiopoietin-2/Tie2 Pathway REACTOME_TIE2_SIGNALING, ANGPT2 axis	Upregulated (during escape)	Resistant – Vascular remodeling & immune evasion	Angiopoietin-2 (ANG2) is often upregulated as VEGF is inhibited ²⁹ . High Ang2 destabilizes vessels but also recruits Tie2-expressing macrophages, facilitating vessel regrowth and immune suppression . Clinical data: patients with elevated Ang2 had poor bevacizumab responses ²⁸ . Bevacizumab temporarily lowers Ang2+ vessel density ¹⁰ , but if tumor Ang2 rebounds or remains high, it creates alternative angiogenic loops and protects endothelial cells from VEGF withdrawal. Thus, Ang2 up = a mechanism of resistance (rationale for combining Ang2 inhibitors).	2

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	Source
HGF/c-MET Pathway REACTOME_SIGNALING_BY_MET, KEGG_HSA04010	Upregulated (under hypoxia)	Resistant – Invasion and proliferation	VEGF blockade-induced hypoxia upregulates HGF and c-MET on tumor cells ³⁰ ³¹ . Active MET signaling drives tumor cell migration (scattering to find blood supply) and growth via MAPK/PI3K. In resistant tumors, c-MET is often overexpressed ; e.g., bevacizumab-treated GBM and aggressive breast tumors had higher MET at progression ³² . MET upregulation allows tumors to become highly invasive (escaping local nutrient deprivation) and can also promote new vessel growth through MET-VEGF cross-talk. Blocking MET has shown to counteract this resistance in other cancers ⁴⁹ .	3

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	Source
Immune Infiltration / T-cell Activation IMMUNOSIGDB_T-CELL_INFLAMMATORY_SIGNATURE	Upregulated (by vessel normalization)	Sensitive – Enhances antitumor effect	Bevacizumab can convert the tumor microenvironment to a more immune-active state. Increased infiltration of CD8⁺ T-cells and mature dendritic cells was observed after one bevacizumab dose in TNBC ¹⁰ . This upregulation of immune pathways (e.g., IFN- γ response, DC maturation) correlates with better tumor control. Essentially, vessel normalization by bevacizumab alleviates hypoxia and high interstitial pressure, enabling immune cells to enter tumors. Tumors with pre-existing immune-upregulation (TILs, chemokines) respond particularly well, as bevacizumab boosts an ongoing immune response. This pathway's upregulation synergizes with therapies like checkpoint inhibitors.	1

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	Source
IL-6/STAT3 Inflammatory Pathway GO_BP_INFLAMMATORY_RESPONSE (IL6-driven)	Upregulated (in certain microenvironments)	Resistant – Pro-angiogenic inflammation	Interleukin-6 and downstream STAT3 are often upregulated in response to VEGF blockade, especially in inflammatory or obese environments ³³ ¹⁹ . IL-6 promotes angiogenesis (via STAT3-induced VEGF/FGF) and recruits immunosuppressive myeloid cells. Patients with high IL-6 (e.g., obese TNBC patients) had larger tumors and weaker bevacizumab efficacy, as IL-6-driven pathways sustained angiogenesis and tumor viability despite VEGF inhibition ³³ ¹⁹ . Thus, an upregulated IL-6 pathway represents a resistance mechanism. Blocking IL-6 has been shown to restore sensitivity in preclinical models (combating this inflammation-fueled resistance) ³⁴ ¹⁹ .	3

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	Source
VEGF/Angiogenesis Baseline Dependency MSigDB "Angiogenesis" signature	Upregulated (baseline state)	Sensitive – Predicts strong response	Tumors with high baseline angiogenic activity – exemplified by elevated VEGF-A levels, VEGFR2 expression, and microvessel density – are markedly sensitive to bevacizumab ⁹. A high "angiogenesis signature" (e.g., VEGF, ANGPTL4, PECAM1 upregulated) means the tumor relies on neovascularization for growth. Bevacizumab in such cases yields large tumor regressions and PFS benefit by targeting this dependency. Clinical evidence: in pooled analyses, the subset of patients with VEGF-A^{high} had significantly longer PFS with bevacizumab than VEGF-A^{low} patients ⁸. This pathway essentially denotes VEGF-addiction – its upregulation at baseline makes the tumor exquisitely vulnerable to VEGF withdrawal.	9

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	So
Alternate Angiogenesis Pathways (lack thereof) <i>FGF/PDGF/Ang-2 (low)</i>	<i>Downregulated / low activity</i> (baseline)	Sensitive – No bypass routes	(Inverse pathway entry) – Tumors that have downregulation or low expression of alternative angiogenic factors are more sensitive to bevacizumab. For example, when FGF2 and ANG2 pathways are not active (down), the tumor cannot compensate for VEGF blockade ²⁶ ⁸³. Similarly, low PIGF or PDGF means VEGF is the sole driver. The absence of parallel angiogenic signals leaves no “escape hatch,” so bevacizumab is highly effective. This was seen in biomarker studies: patients whose tumors had low baseline FGF2 had improved responses to bevacizumab ¹⁹. Thus, downregulation of redundant angiogenesis = heightened sensitivity, as the tumor is essentially one-track (VEGF-dependent).	2

Pathway (ID/Name)	Regulation	Effect of Bevacizumab	Biological Rationale	So
Angiogenic Dormancy (vessel co-option) <i>
Downregulated Angiogenesis</i>	Downregulated (intrinsically low angiogenesis)	Resistant – Intrinsic non-response	Some breast tumors exhibit low angiogenic profiles (minimal VEGF, vessels) and grow via vessel co-option or not at all (“dormant” vasculature) ¹⁶ . This downregulation of angiogenic signaling confers innate resistance to bevacizumab – there is little for the drug to target. Clinically, slow-growing luminal tumors or heavily pretreated lesions often have low microvessel density and derive negligible benefit from anti-VEGF ¹⁶ . The tumor survives by relying on existing blood supply or low oxygen demand. In such cases, bevacizumab’s pathway (angiogenesis) is moot; the tumor’s angiogenesis-off state means the drug yields no advantage (and these patients were the non-responders in trials).	1

Sources: Key evidence from clinical trials, biomarker studies, and pathway databases has been cited. For example, DrugBank/ChEMBL confirm bevacizumab's target and mechanism, PubMed studies detail the hypoxia and EMT changes with resistance ¹⁷ ¹⁸, and MSigDB/Reactome identifiers are used to link pathways to their human gene sets. Clinical guidelines (FDA/EMA, NCCN, ESMO) and trials (E2100, AVADO, etc.) substantiate the relationships between these pathways and bevacizumab sensitivity or resistance in breast cancer. Each pathway listed was included based on breast cancer-specific data or strong mechanistic rationale in human models, per the structured research brief.

¹ ¹¹ Bevacizumab - Wikipedia

<https://en.wikipedia.org/wiki/Bevacizumab>

² ³ ⁴ ⁶ ⁸ ⁵¹ ⁵⁶ First-line bevacizumab in combination with chemotherapy for HER2-negative metastatic breast cancer: pooled and subgroup analyses of data from 2447 patients - Annals of Oncology

[https://www.annalsofoncology.org/article/S0923-7534\(19\)37346-6/fulltext](https://www.annalsofoncology.org/article/S0923-7534(19)37346-6/fulltext)

⁵ ⁷⁶ ⁷⁷ ⁷⁸ ⁷⁹ Avastin OD WD Memo 12.15.10.FINAL.doc

<https://www.fda.gov/media/79525/download>

⁷ SEOM-GEICAM-SOLTI clinical guidelines in advanced breast cancer (2022) | Clinical and Translational Oncology

<https://link.springer.com/article/10.1007/s12094-023-03203-8>

⁹ Biomarker results from the AVADO phase 3 trial of first ... - PubMed

<https://pubmed.ncbi.nlm.nih.gov/23422754/>

¹⁰ ²⁹ ⁸² Bevacizumab improves tumor infiltration of mature dendritic cells and effector T-cells in triple-negative breast cancer patients | npj Precision Oncology

https://www.nature.com/articles/s41698-021-00197-w?error=cookies_not_supported&code=e355e22b-a522-4240-aa59-1ba77da2400f

¹² ¹³ ¹⁴ ¹⁵ Bevacizumab: Uses, Interactions, Mechanism of Action | DrugBank Online

<https://go.drugbank.com/drugs/DB00112>

¹⁶ ³⁹ ⁴⁰ ⁴¹ Angiogenesis and lymphangiogenesis are downregulated in primary breast cancer | British Journal of Cancer

https://www.nature.com/articles/6605219?error=cookies_not_supported&code=4da439b0-35b3-4d1c-a81d-5ab6004c313b

¹⁷ ¹⁸ ²⁰ ²¹ ²² ²⁶ ²⁷ ³⁰ ³¹ ³² ³⁵ ³⁶ ³⁷ ³⁸ ⁴⁵ ⁴⁸ ⁴⁹ ⁵⁰ Frontiers | Resistance Mechanisms to Anti-angiogenic Therapies in Cancer

<https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2020.00221/full>

¹⁹ ³³ ³⁴ ⁵² Ludwig Cancer Research

<https://www.ludwigcancerresearch.org/news-releases/obesity-may-promote-resistance-to-antiangiogenic-therapy-for-breast-cancer/>

²³ ²⁴ ²⁵ ⁴² ⁴⁴ ⁵⁴ Gene Expression Analysis Reveals Distinct Pathways of Resistance to Bevacizumab in Xenograft Models of Human ER-Positive Breast Cancer

<https://scholarworks.indianapolis.iu.edu/items/61ba28d2-87a5-4606-a4ad-3ee2dd93e4b6>

²⁸ Resistance to Anti-Angiogenic Therapy in Cancer—Alterations ... - NIH

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5979390/>

43 An update on biomarkers of potential benefit with bevacizumab for ...

<https://pmc.ncbi.nlm.nih.gov/articles/PMC6736652/>

46 Angiogenic factors FGF2 and PDGF-BB synergistically promote ...

<https://pmc.ncbi.nlm.nih.gov/articles/PMC1994630/>

47 Therapeutic paradigm of dual targeting VEGF and PDGF for ... - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC7382445/>

53 Atezolizumab plus paclitaxel and bevacizumab as first-line treatment ...

<https://www.nature.com/articles/s41591-025-03734-3>

55 75 [PDF] AVASTIN - European Medicines Agency (EMA)

https://www.ema.europa.eu/en/documents/variation-report/avastin-h-c-582-ii-0025-epar-assessment-report-variation_en.pdf

57 58 60 61 62 63 64 65 67 68 70 71 72 73 AVASTIN® (bevacizumab) Label

https://www.accessdata.fda.gov/drugsatfda_docs/label/2009/125085s0169lbl.pdf

59 Bevacizumab (Avastin®) in cancer treatment: A review of 15 years of ...

<https://www.sciencedirect.com/science/article/pii/S0305737220300554>

66 Avastin® (bevacizumab) Important Safety Information

<https://www.avastin.com/hcp/safety.html>

69 Contraindicated Use of Bevacizumab and Toxicity in Elderly Patients ...

<https://ascopubs.org/doi/10.1200/JCO.2012.48.4857>

74 81 ABC5 International Consensus Conference on ... - Thieme Connect

<https://www.thieme-connect.com/products/ejournals/pdf/10.1055/a-1139-9380.pdf>

80 Bevacizumab - ESMO-MCBS Scorecards | ESMO

<https://www.esmo.org/guidelines/esmo-mcbs/esmo-mcbs-for-solid-tumours/esmo-mcbs-scorecards/scorecard-363-1>

83 Targeting Angiogenesis in Breast Cancer: Current Evidence and ...

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8913593/>